Weekly Report

Dean Reiter

TRIUMF DarkLight Group Meeting

June 30, 2026

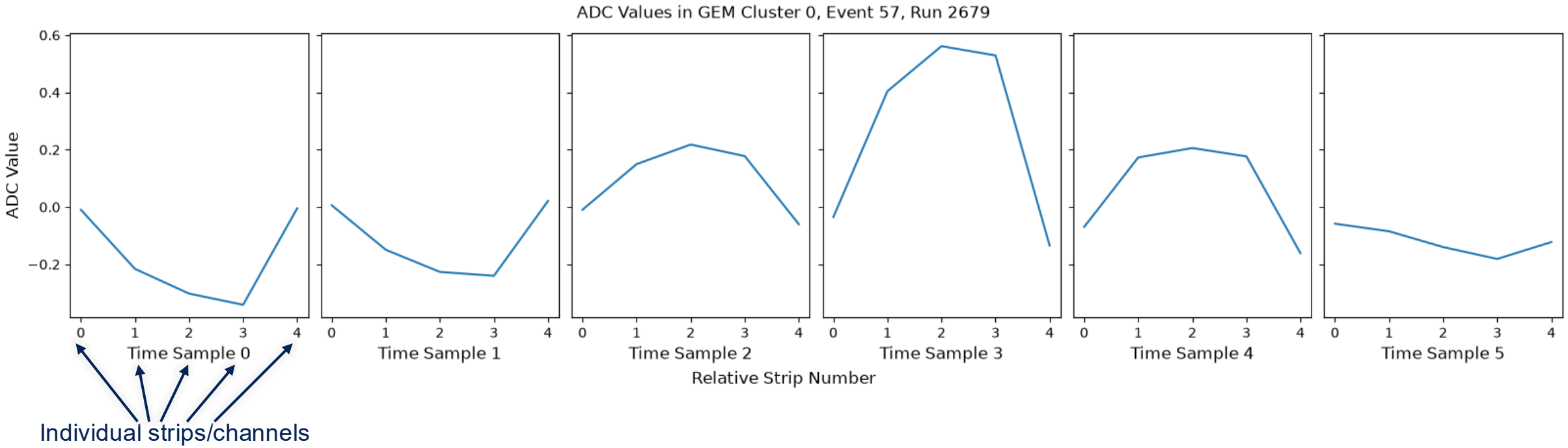
Summary

1. Calculated approximate GEM cluster times
2. Tried looking for correlation between cluster and trigger times in particle tracks
3. Need to review GEMini code and consider next steps

Example of a cluster

Pedestal-subtracted values across five strips over six time samples

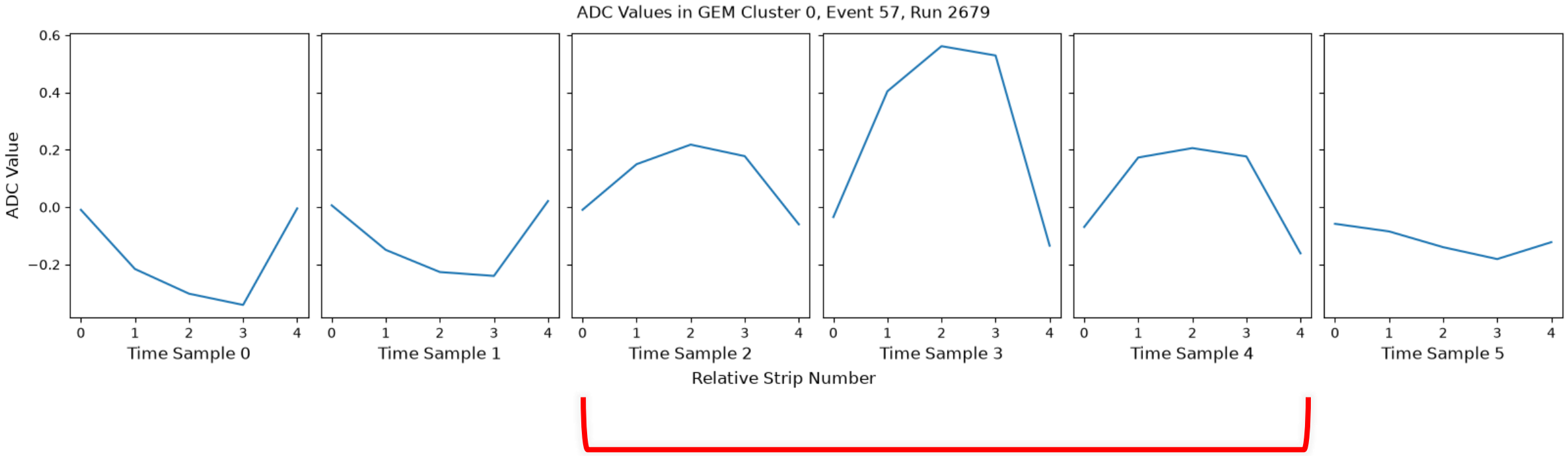
These five strips were identified as containing a cluster in this event



Example of a cluster

Pedestal-subtracted values across five strips over six time samples

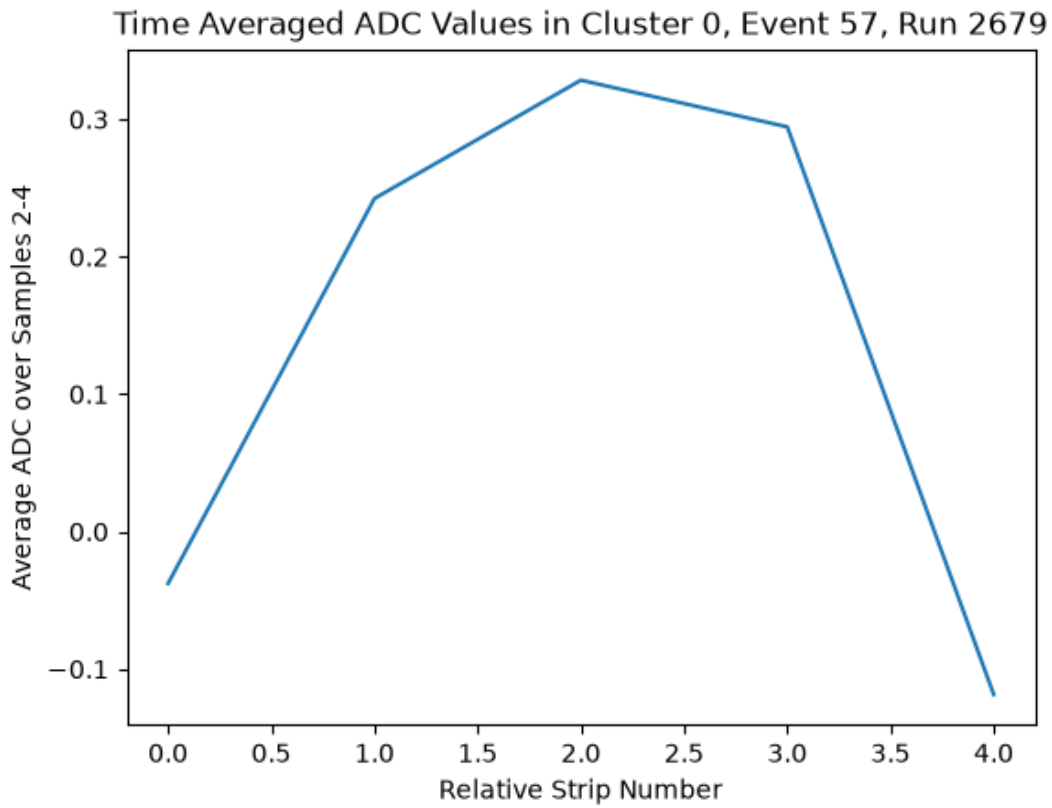
These five strips were identified as containing a cluster in this event



The code takes the average of samples 2 through 4 before searching for clusters

Example of a cluster

This is the time-averaged data which Andy's algorithm identified as a cluster

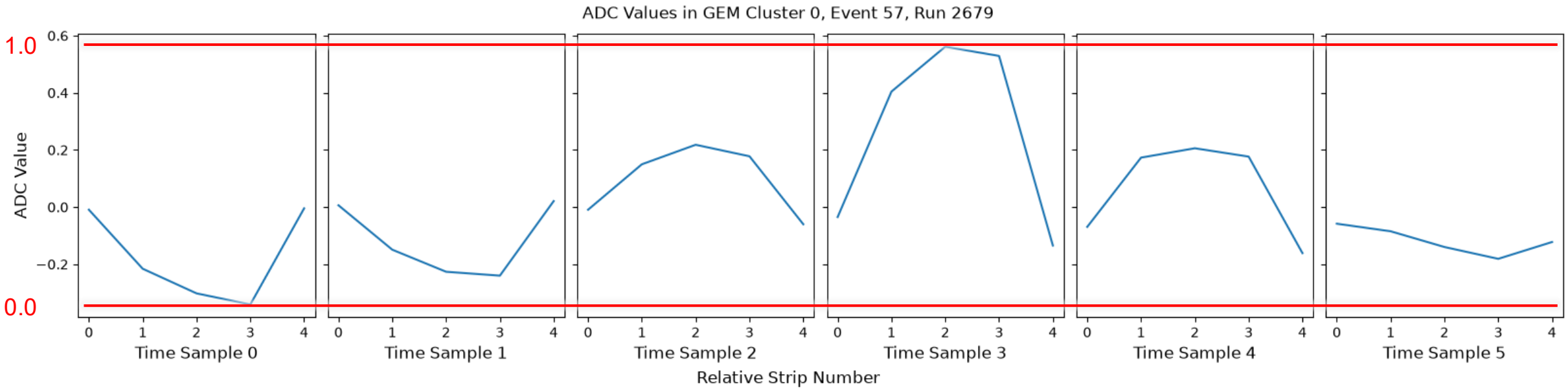


Approximating Cluster Time

Two rough methods of determining the time of a cluster:

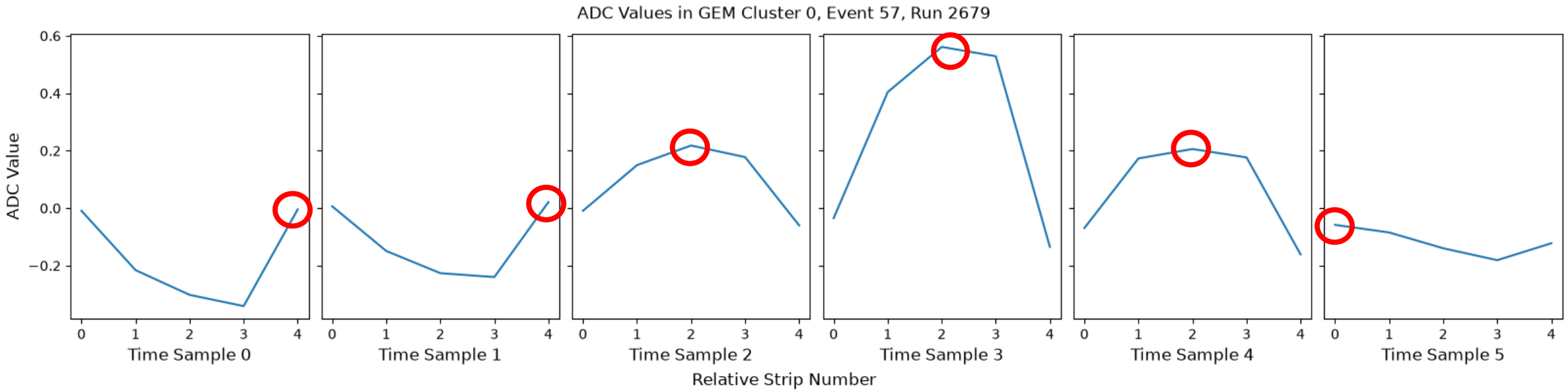
1. “ADC Time”
2. ”Charge Time”

Approximating Cluster Time – “ADC Time”



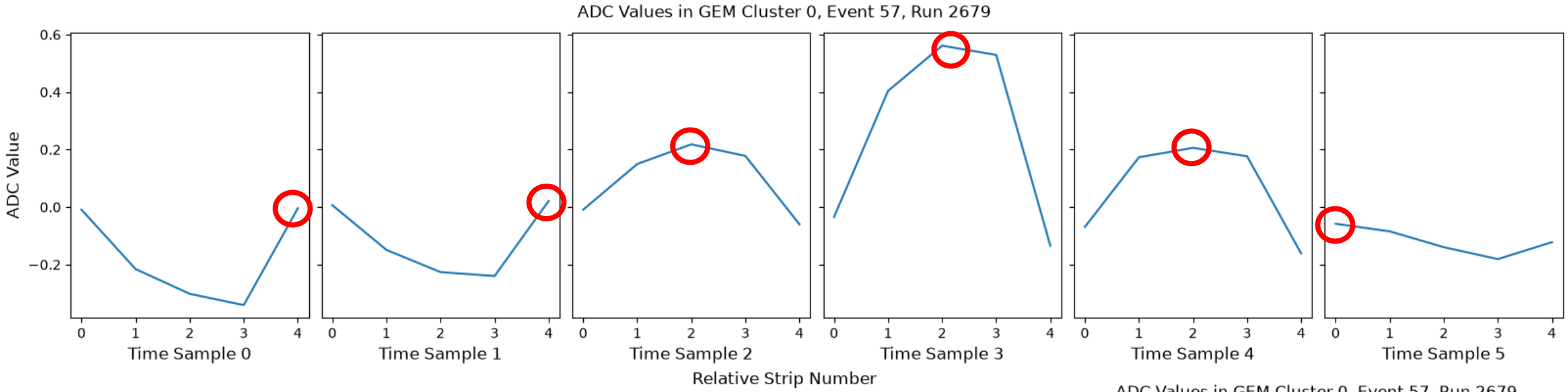
1. Normalize all values to the overall minimum and maximum

Approximating Cluster Time – “ADC Time”

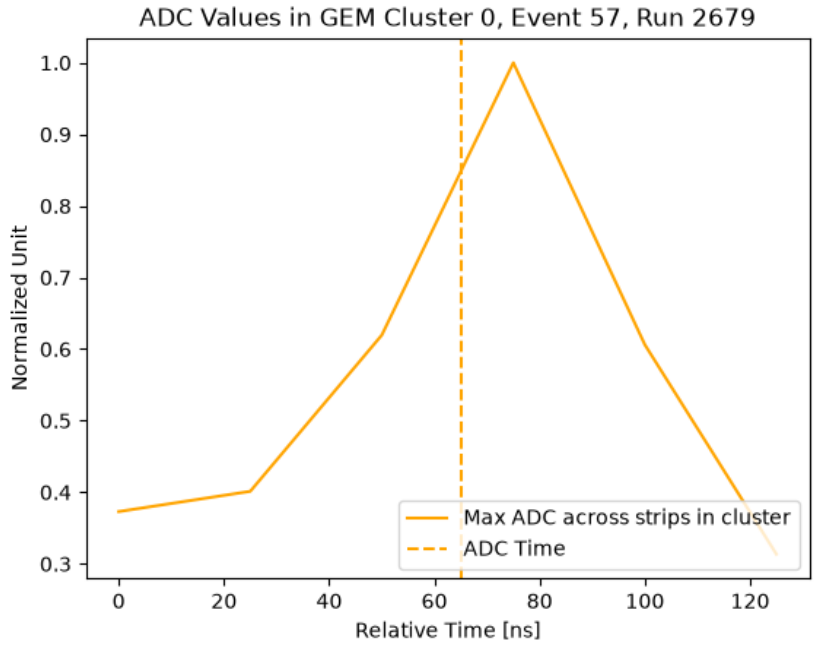


1. Normalize all values to the overall minimum and maximum
2. Find maximum channel value in each sample

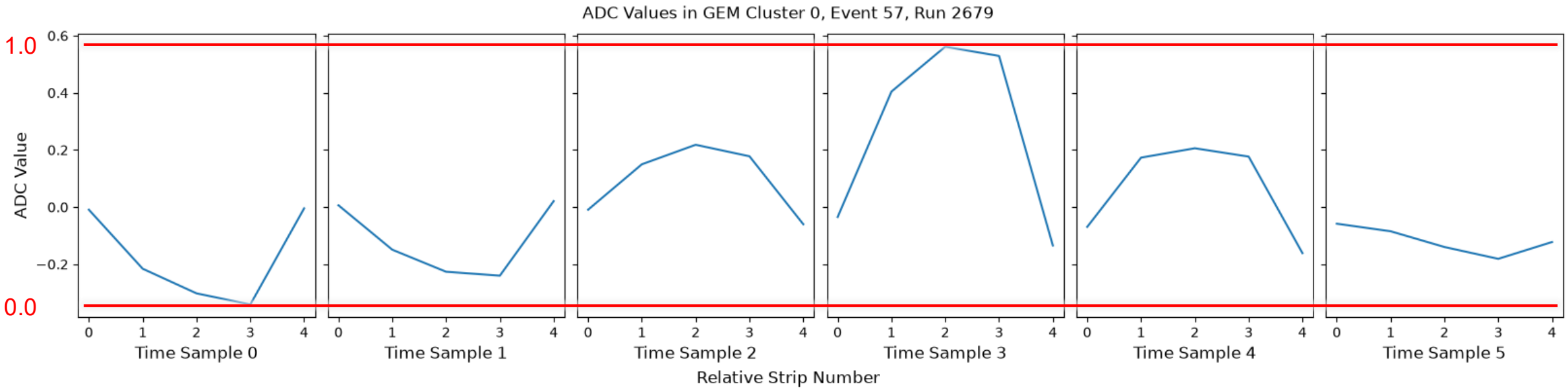
Approximating Cluster Time – “ADC Time”



1. Normalize all values to the overall minimum and maximum
2. Find maximum channel value in each sample
3. Calculate ”mean” time weighted by these maximum values

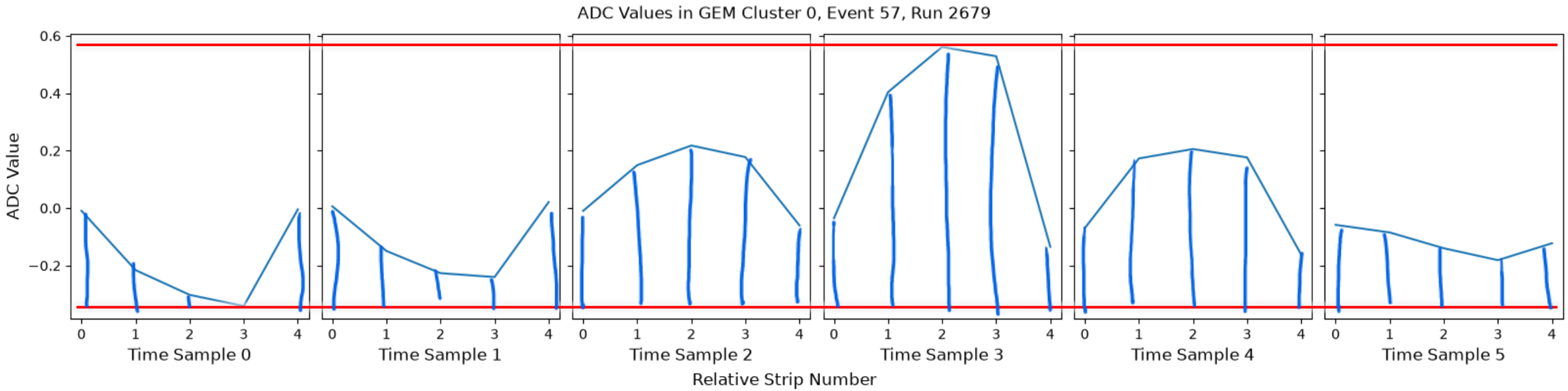


Approximating Cluster Time – “Charge Time”



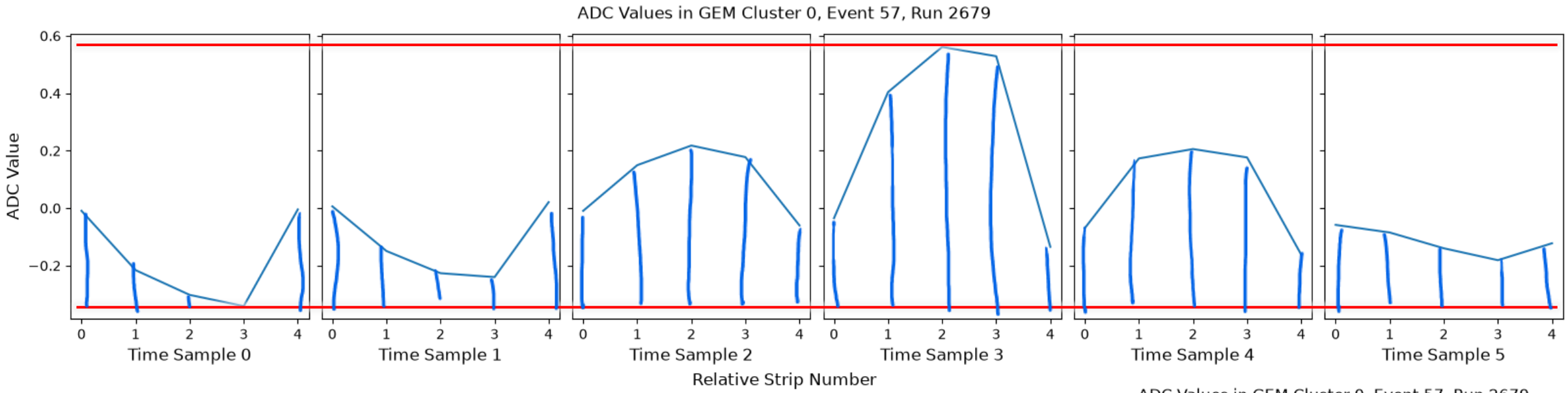
1. Normalize all values to the overall minimum and maximum

Approximating Cluster Time – “Charge Time”

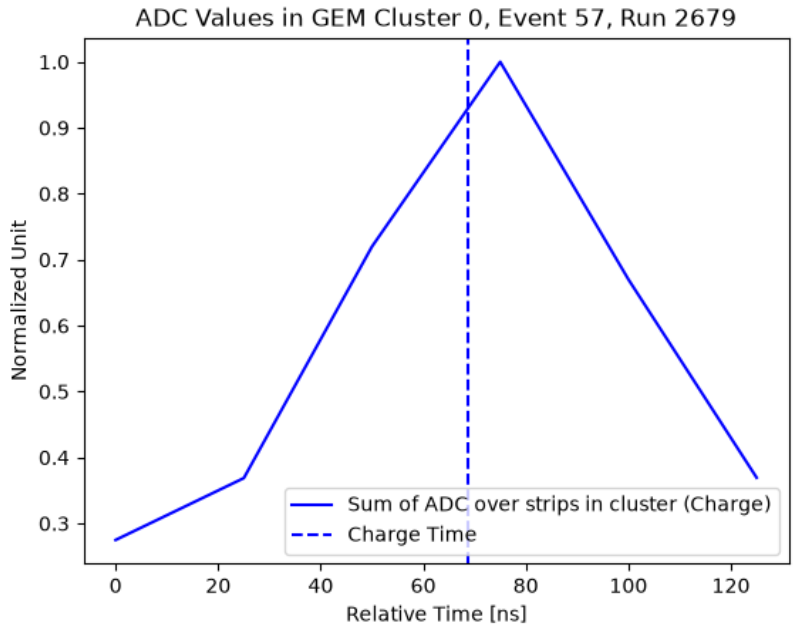


1. Normalize all values to the overall minimum and maximum
2. Calculate “charge” in each sample = sum of channel values

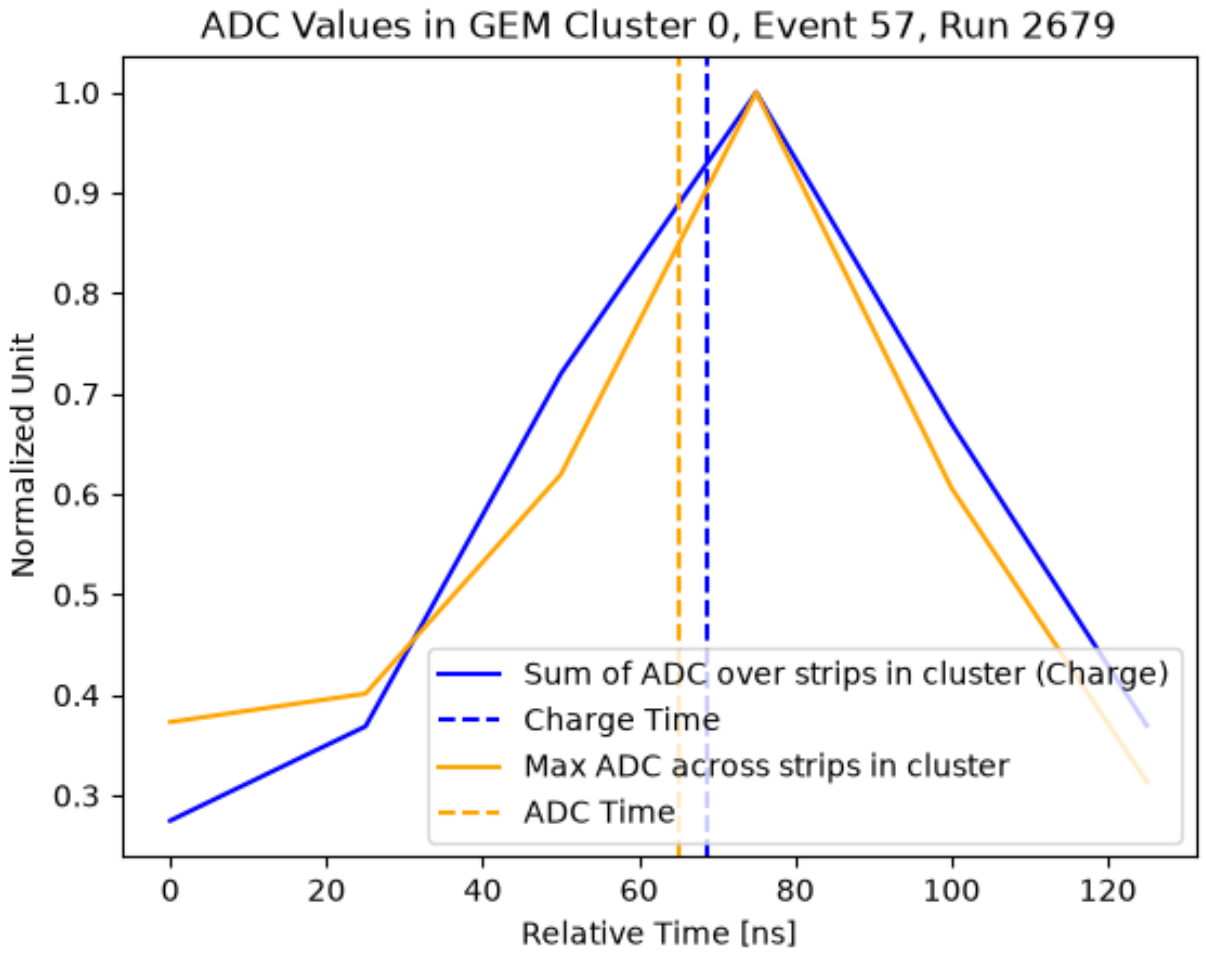
Approximating Cluster Time – “Charge Time”



1. Normalize all values to the overall minimum and maximum
2. Calculate “charge” in each sample = sum of channel values
3. Calculate ”mean” time weighted by these charge values



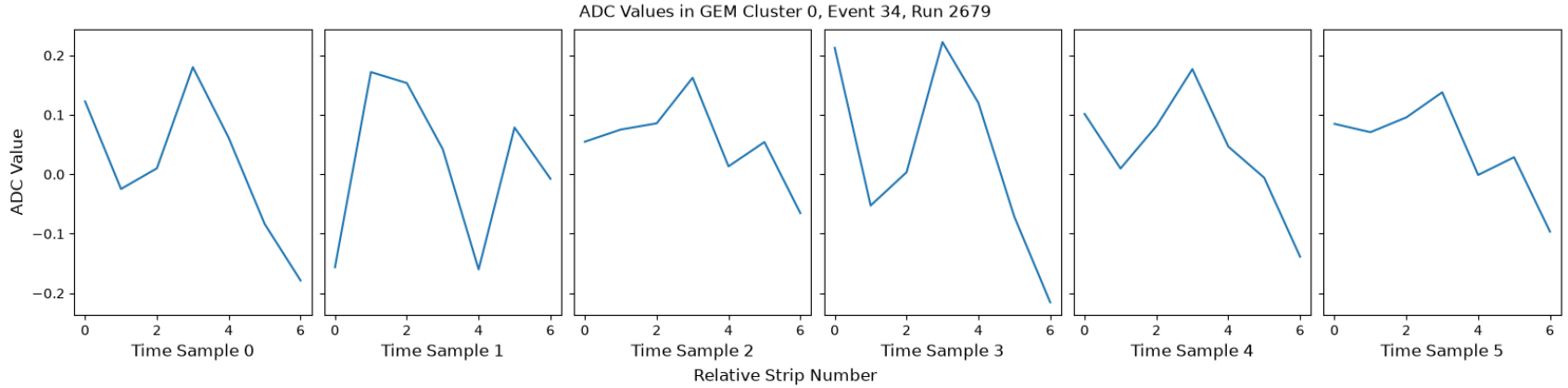
Approximating Cluster Time

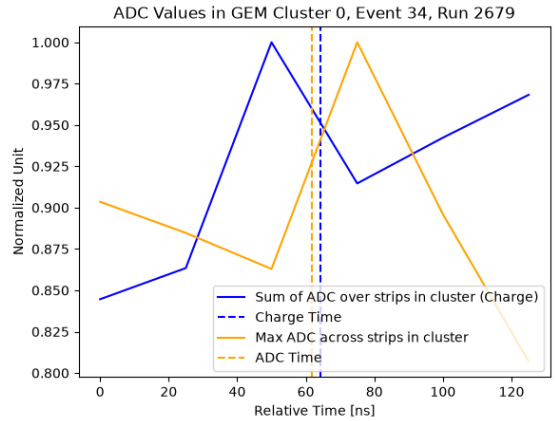


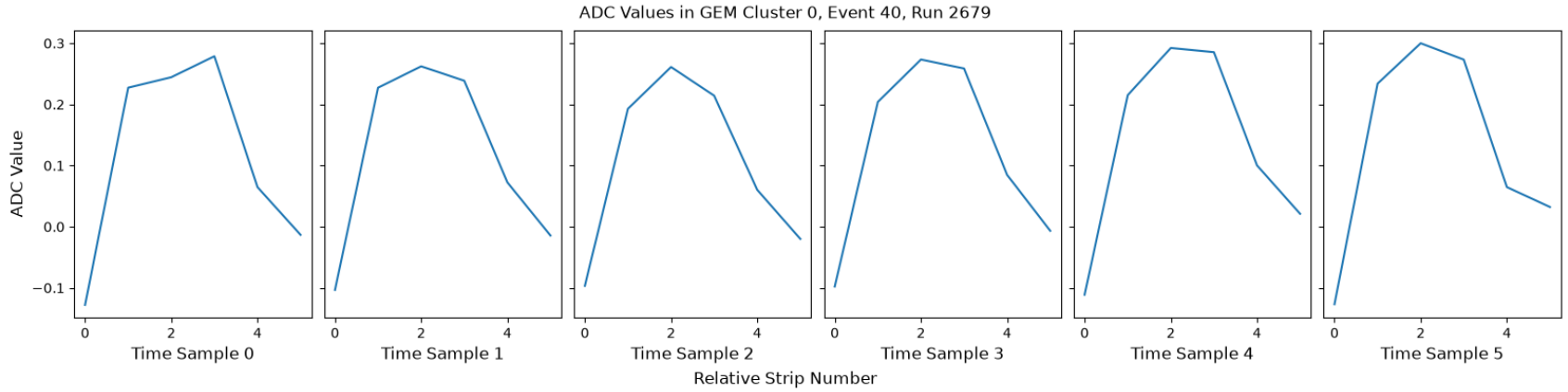
Approximating Cluster Time – Issues

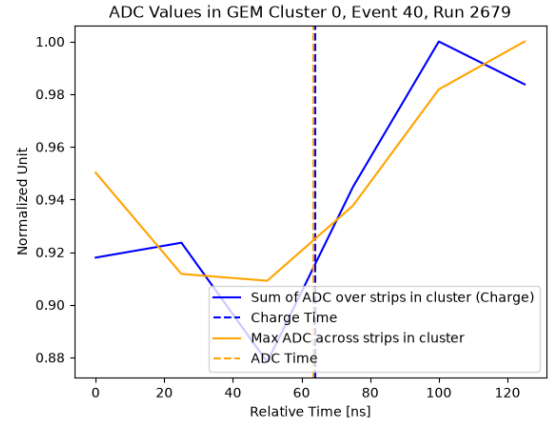
- The slides I presented in the software meeting last week were all wrong
  - Made an error in how I indexed through the data
- These two methods are really bad
  - They both seem to bias the time to the middle of the time window (62.5ns)
  - Should do a fit
- Many clusters are not ideal
  - A lot of noise in gem data
  - Andy's algorithm should be tested thoroughly with real data

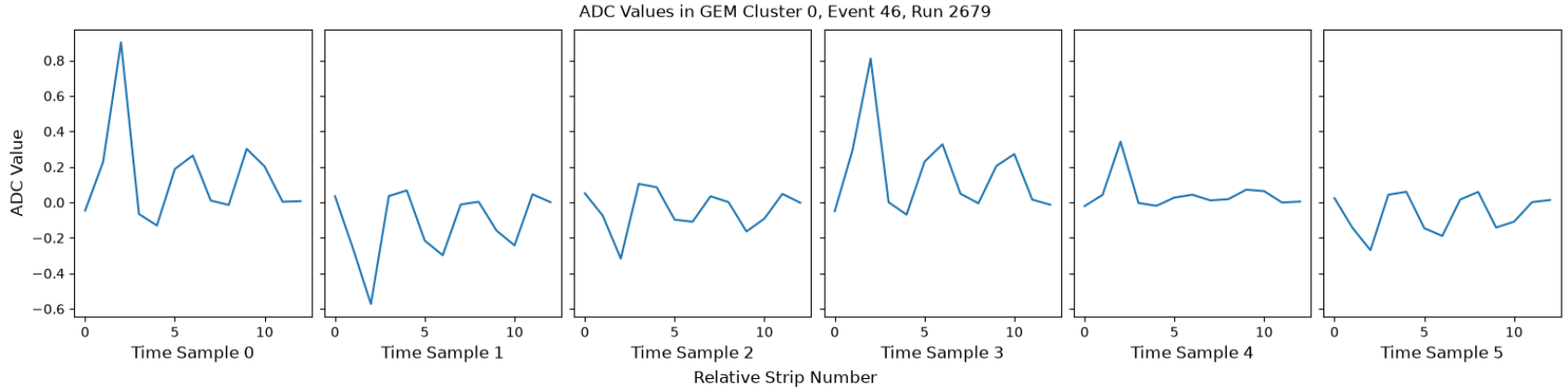
Other examples of clusters

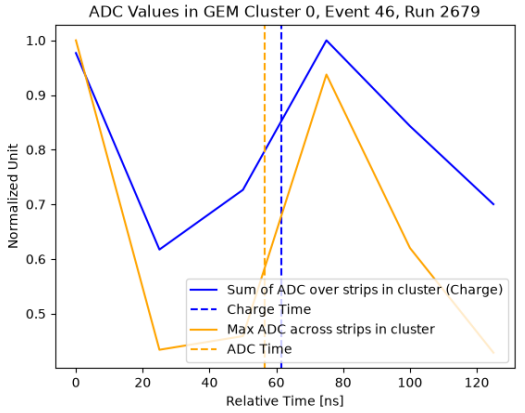










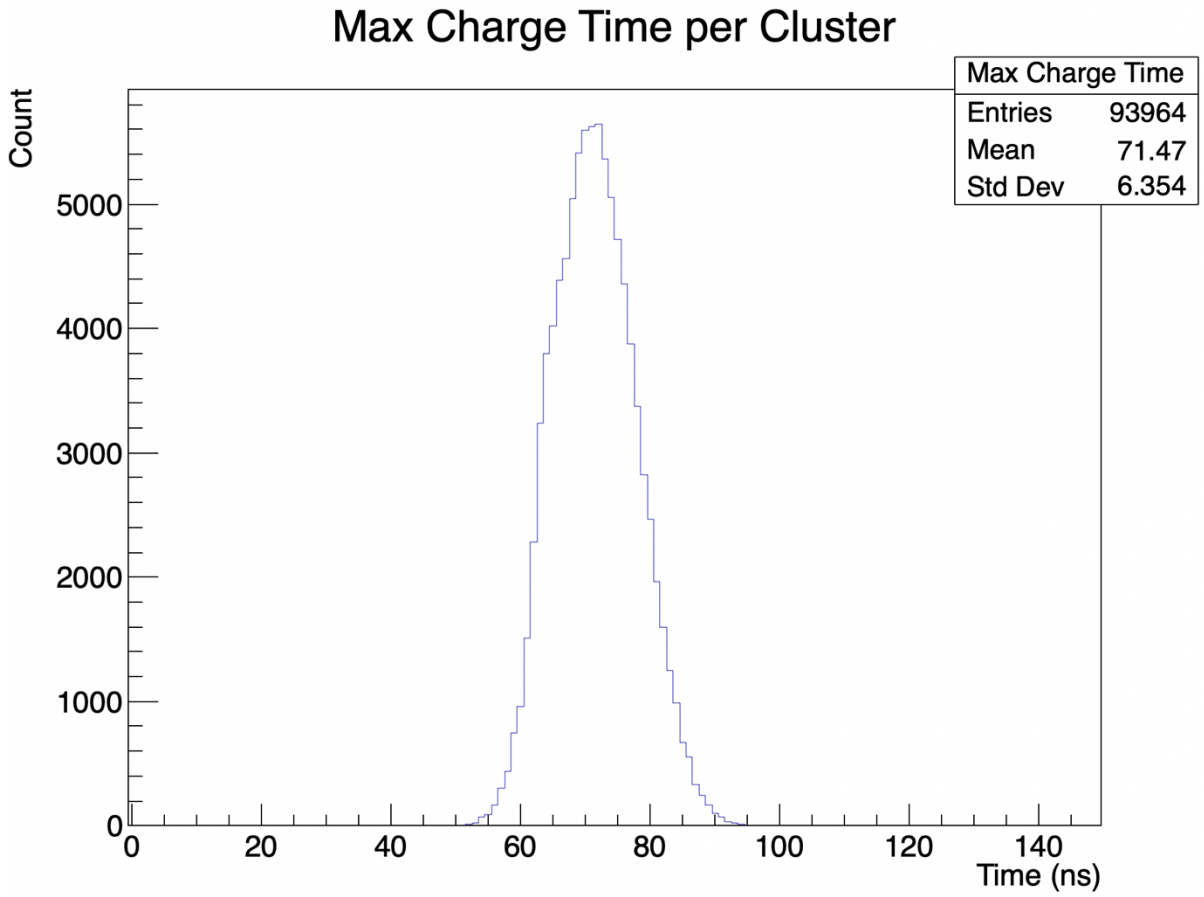


Cluster Time Distribution

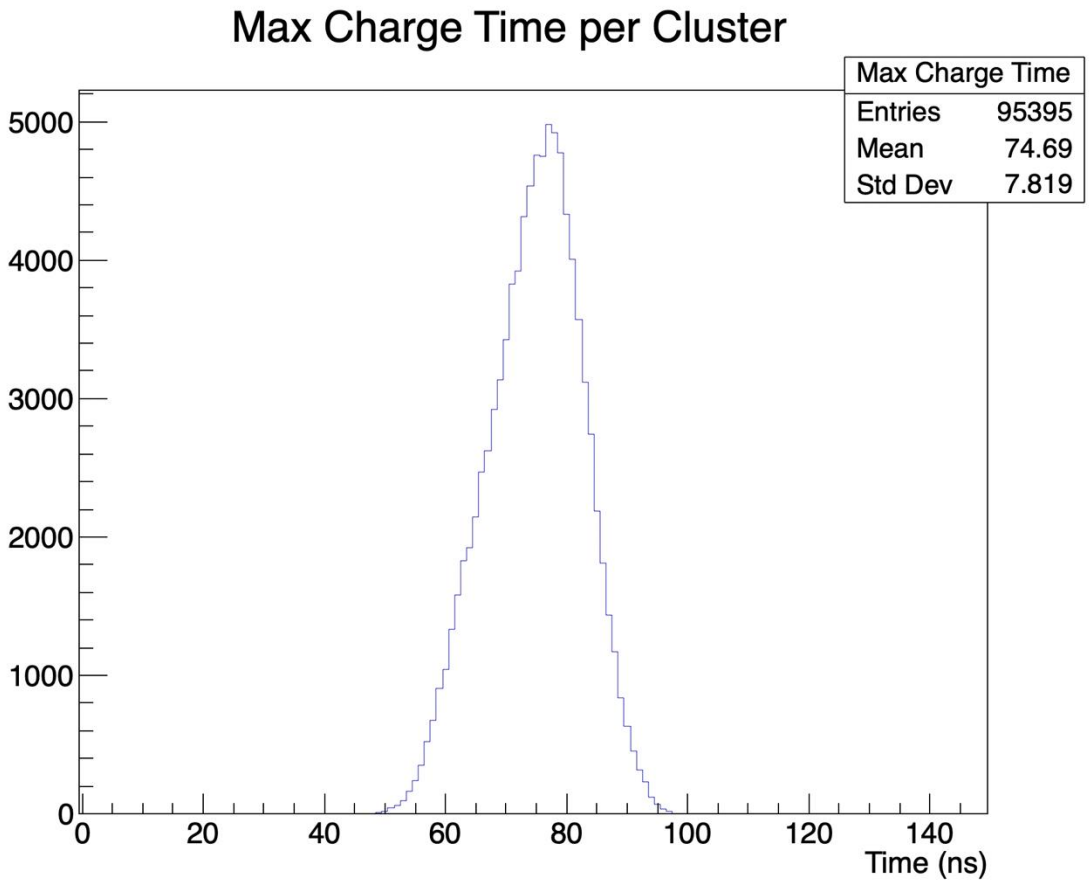
Charge Time

Left Top GEM clusters

Run 3622 - Cosmic



X axis



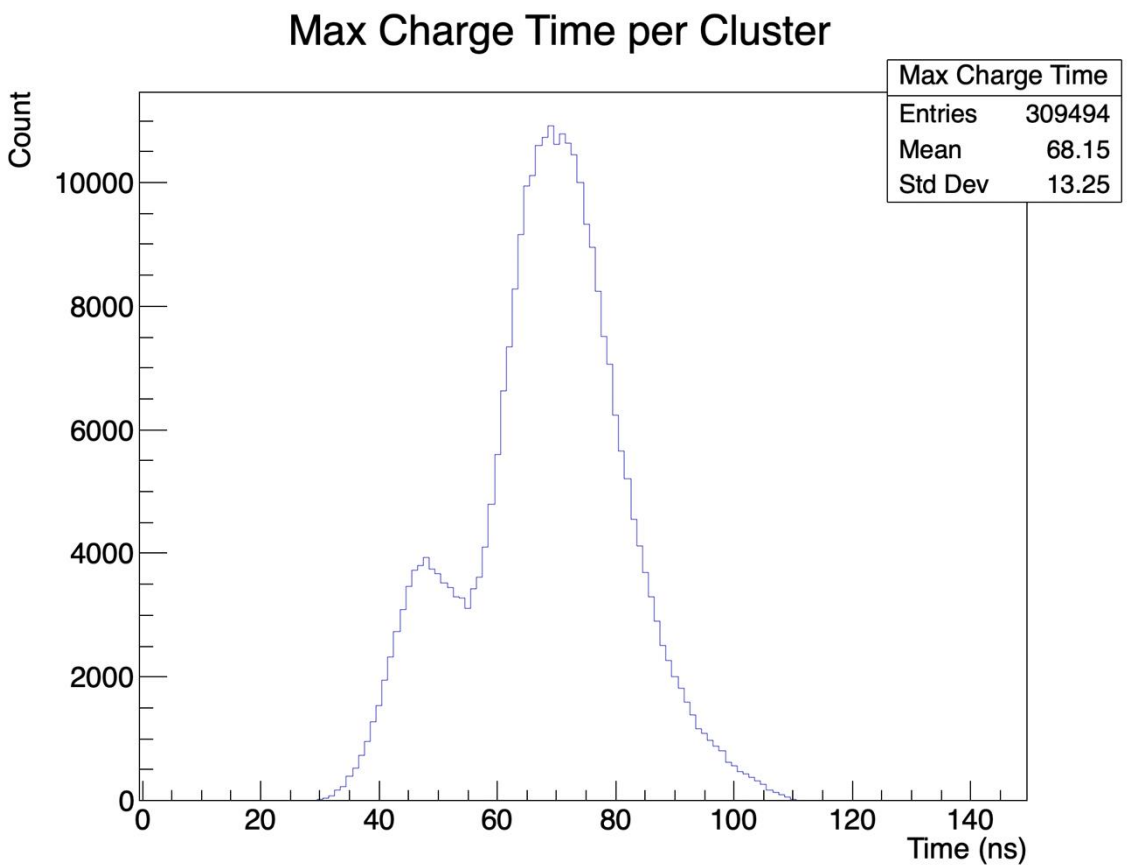
Y axis

Cluster Time Distribution

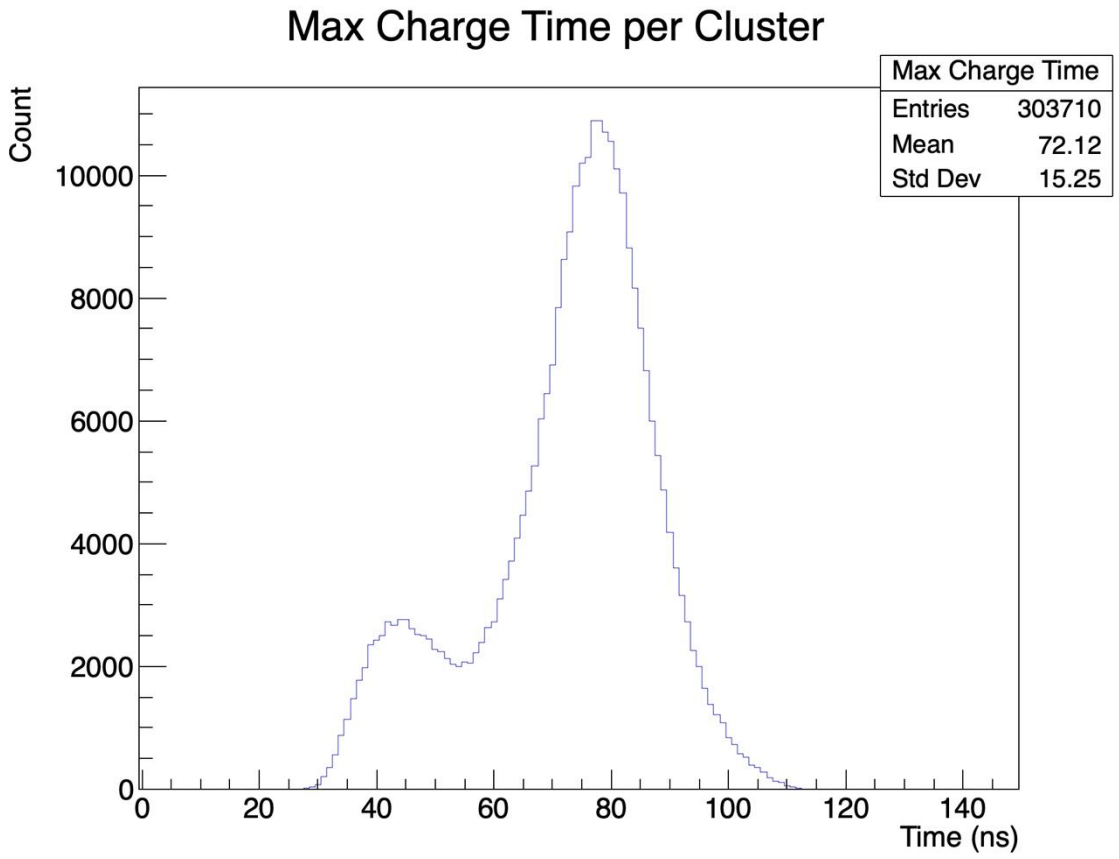
Charge Time

Left Top GEM clusters

Run 4010 - Elastic



X axis



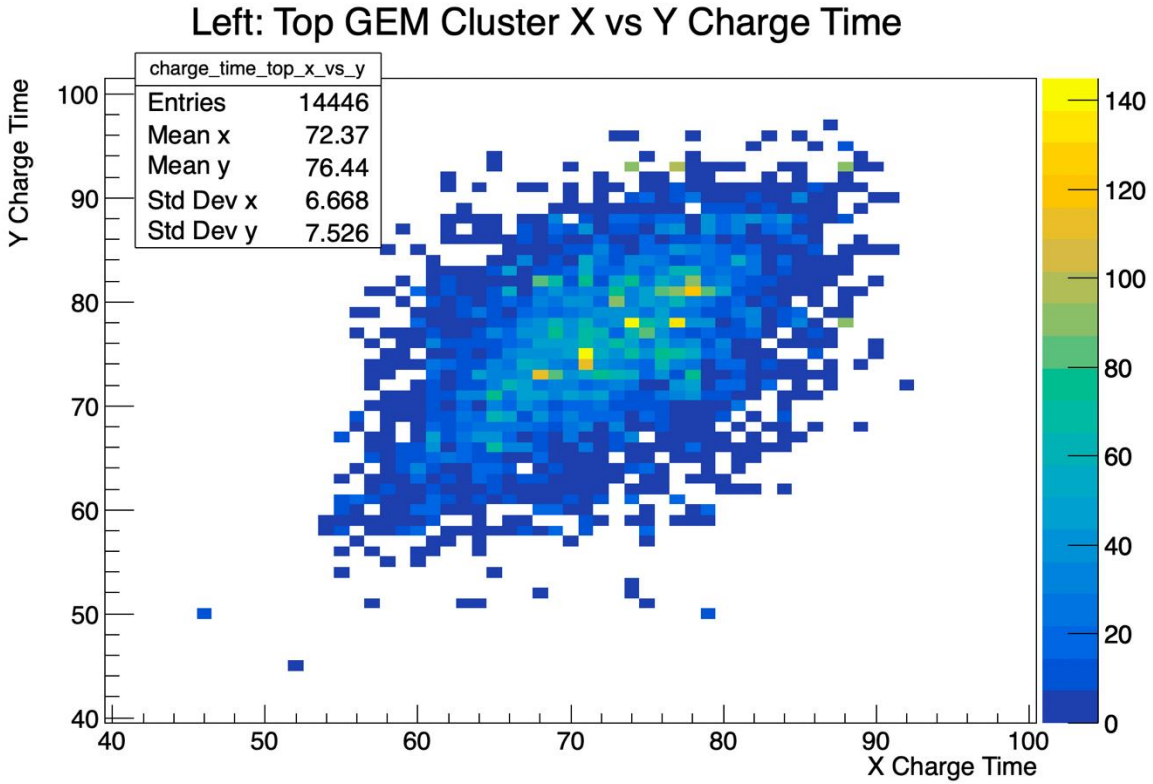
Y axis

Correlation Between X and Y axis Cluster Times

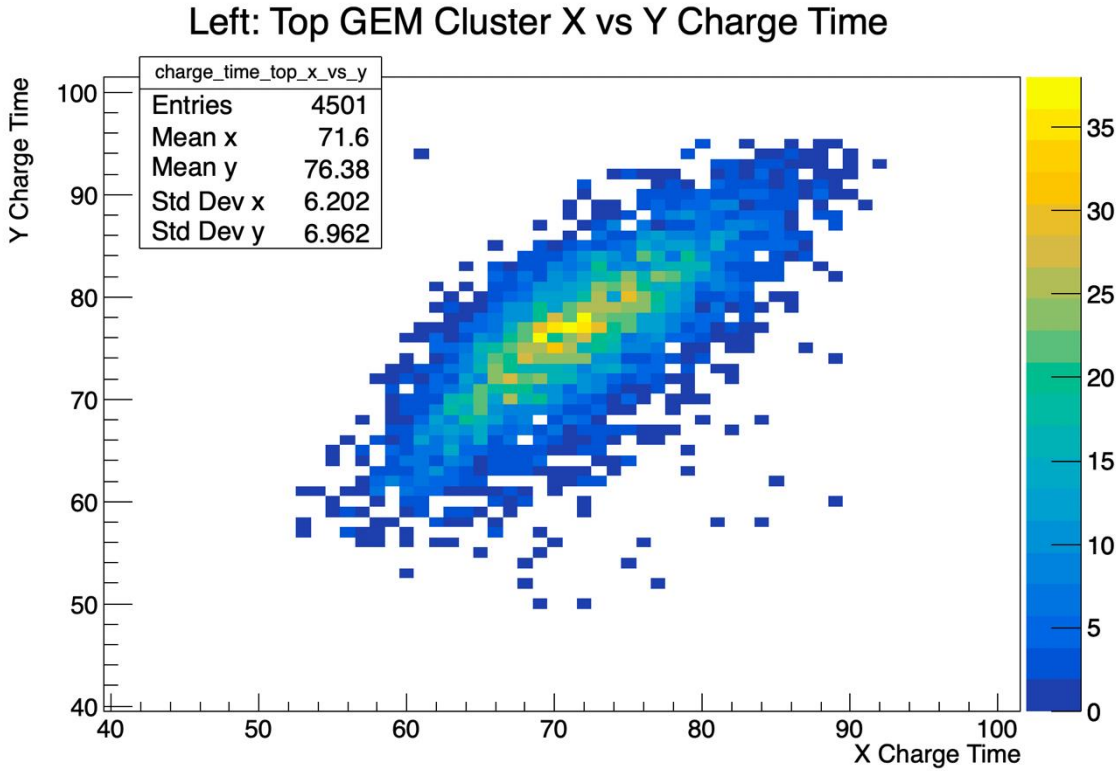
Charge Time

Run 3622 - Cosmic

Left Top GEM clusters



All tracks



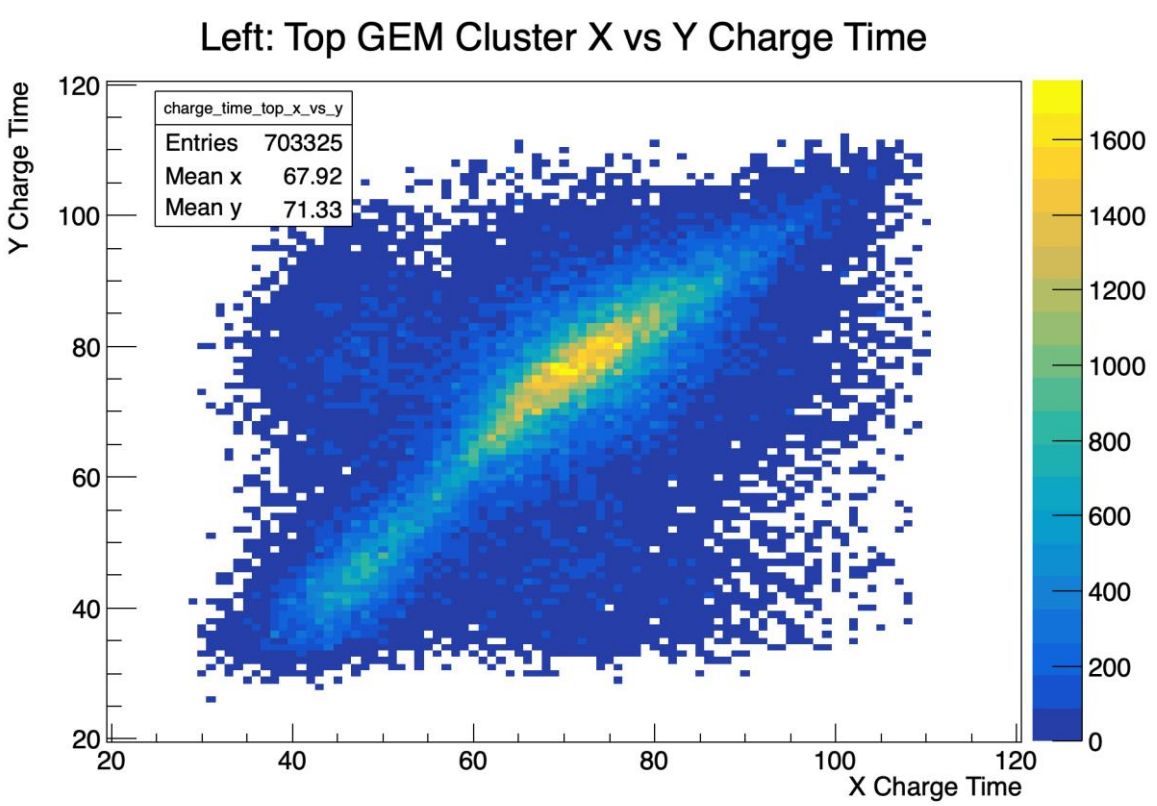
Single (one hit per detector per event) and On-Paddle (hit scintillators) tracks

Correlation Between X and Y axis Cluster Times

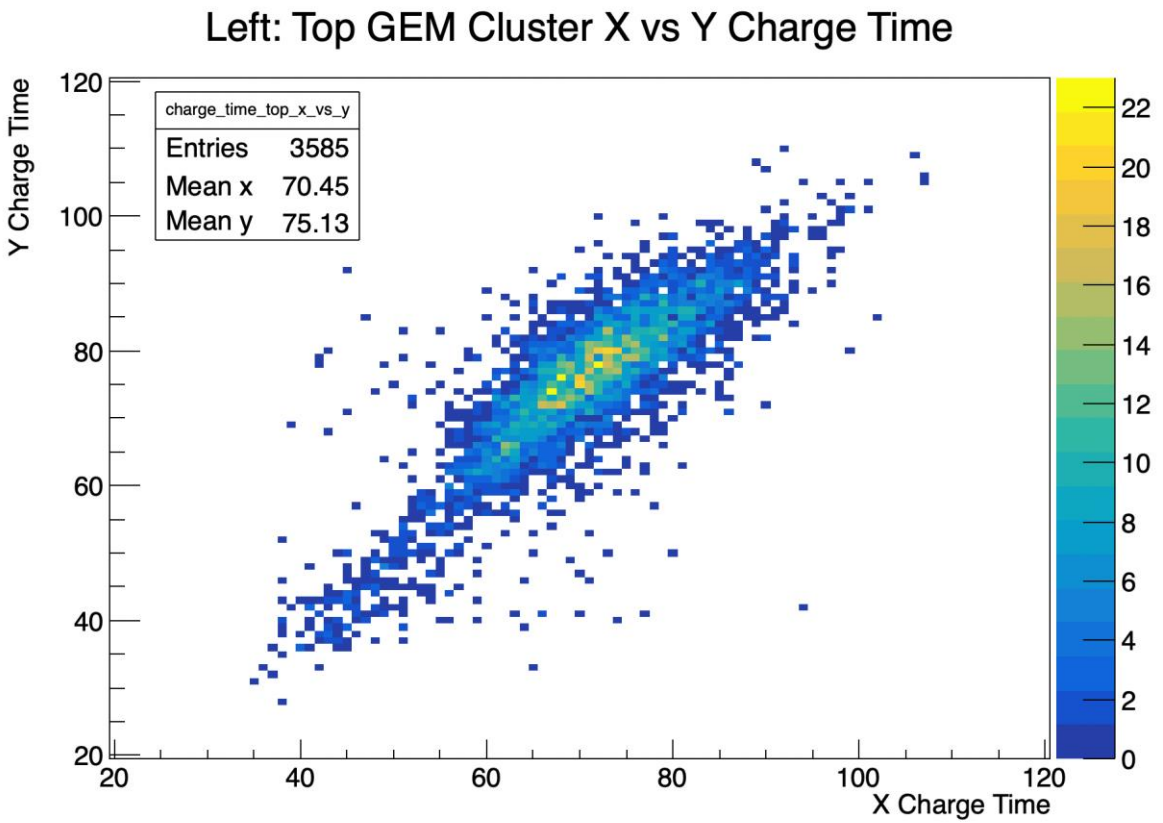
Charge Time

Run 4010 - Elastic

Left Top GEM clusters



All tracks



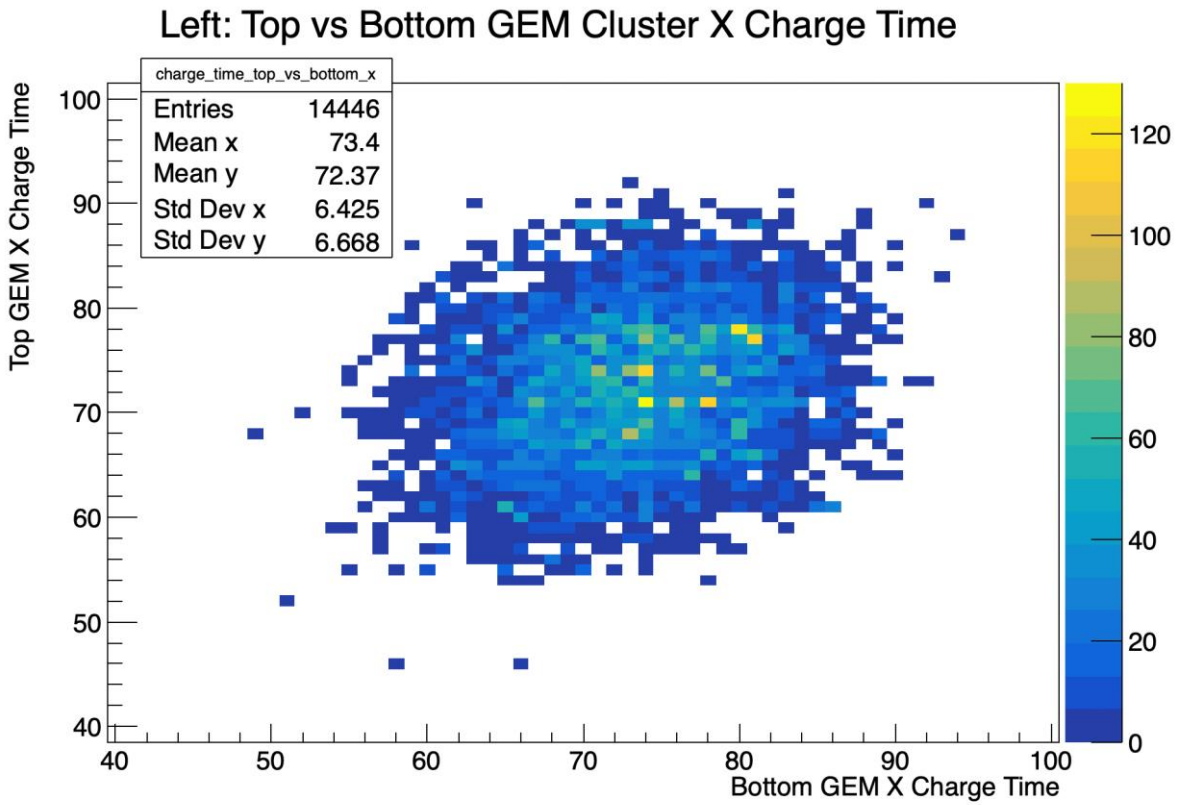
Single (one hit per detector per event) and On-Paddle (hit scintillators) tracks

Correlation Between Bottom and Top GEM Cluster Times

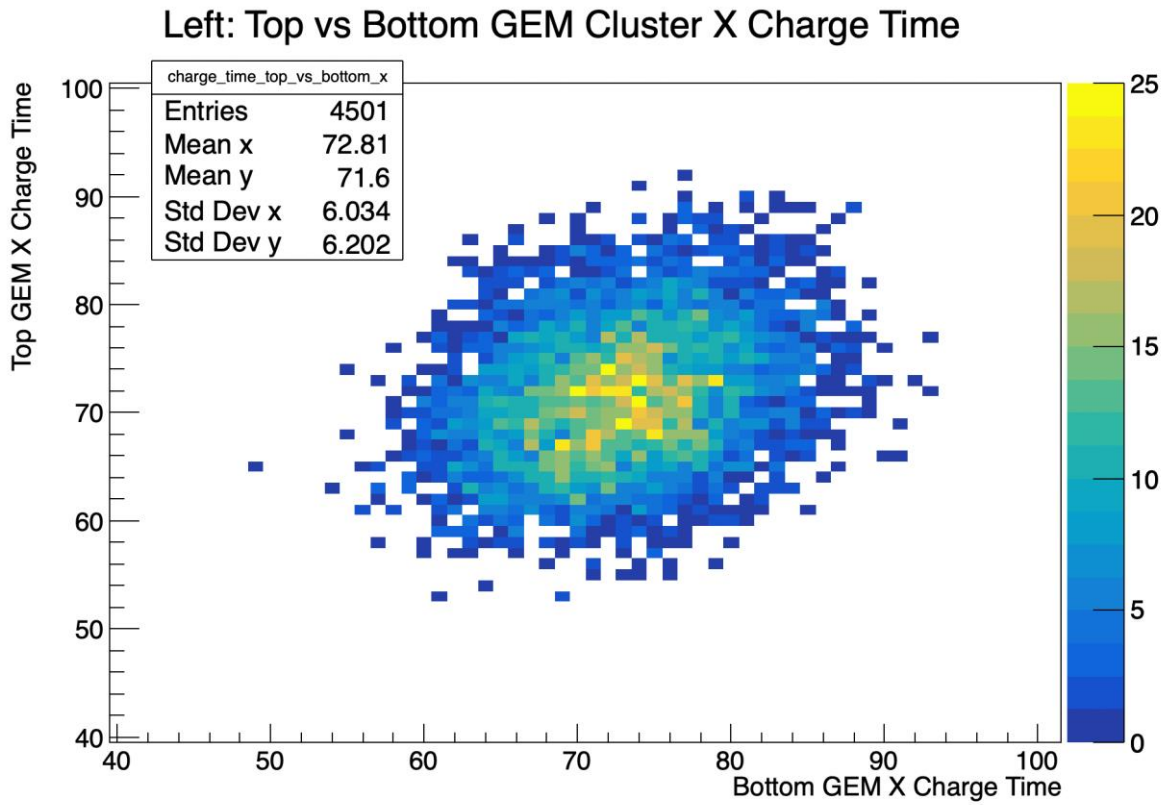
Charge Time

X axis clusters

Run 3622 - Cosmic



All tracks



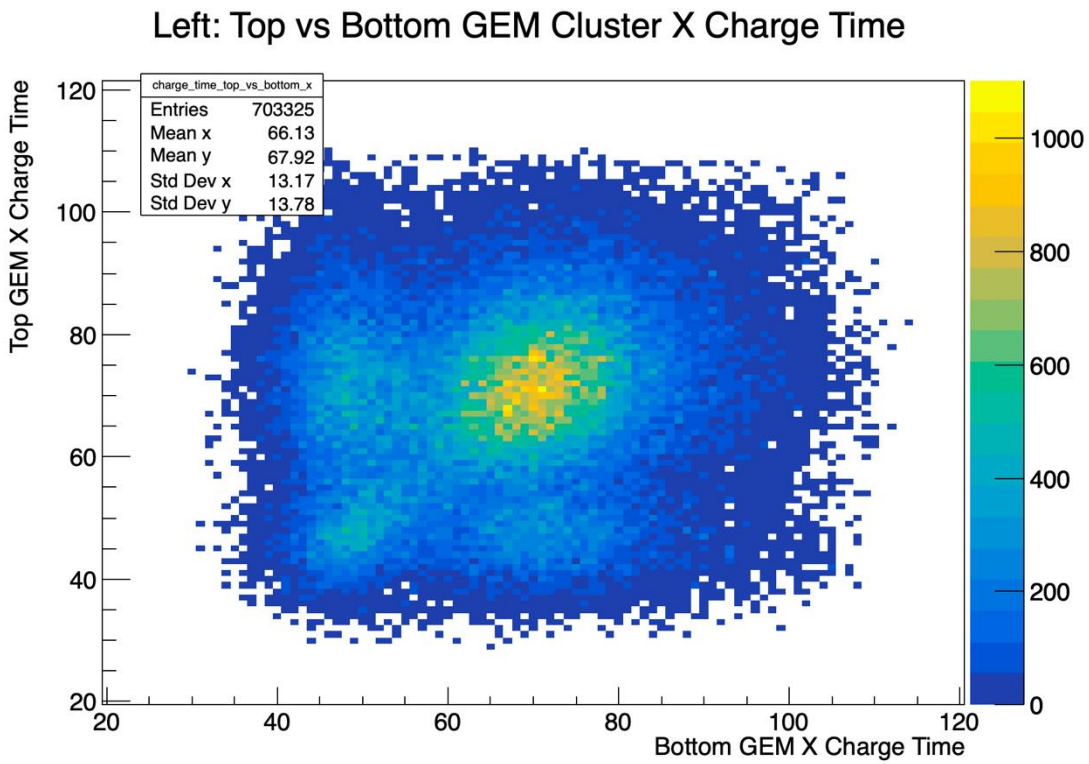
Single (one hit per detector per event) and On-Paddle (hit scintillators) tracks

Correlation Between X and Y axis Cluster Times

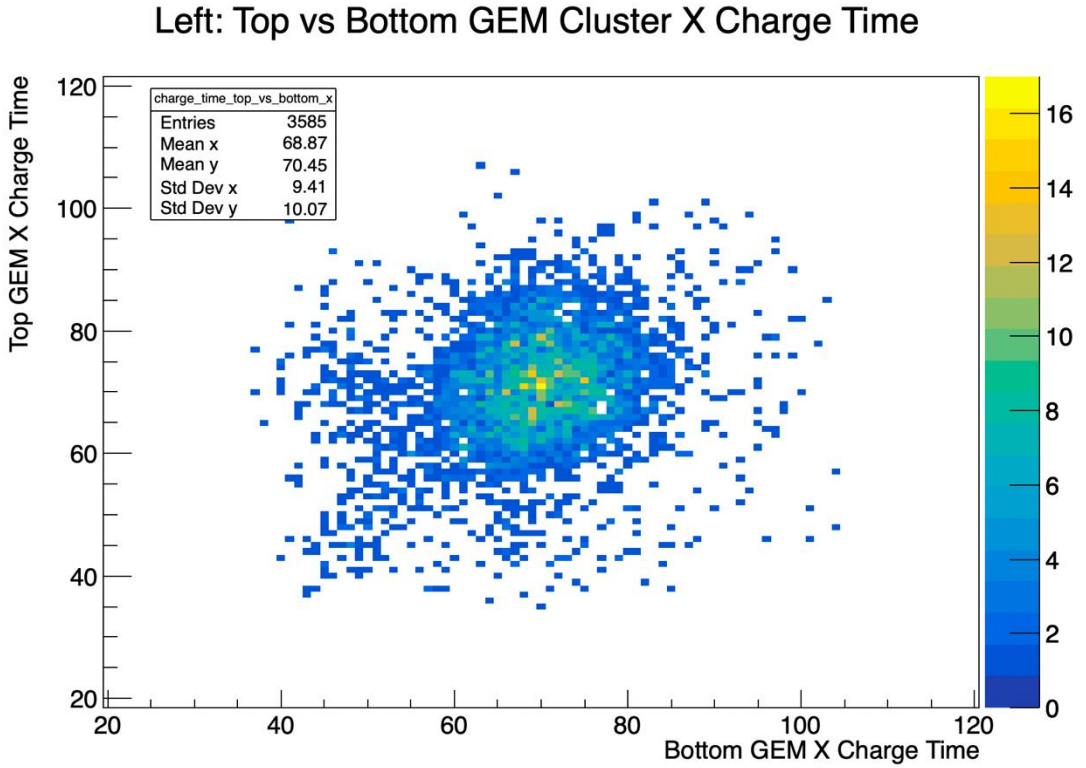
Charge Time

X axis clusters

Run 4010 - Elastic



All tracks



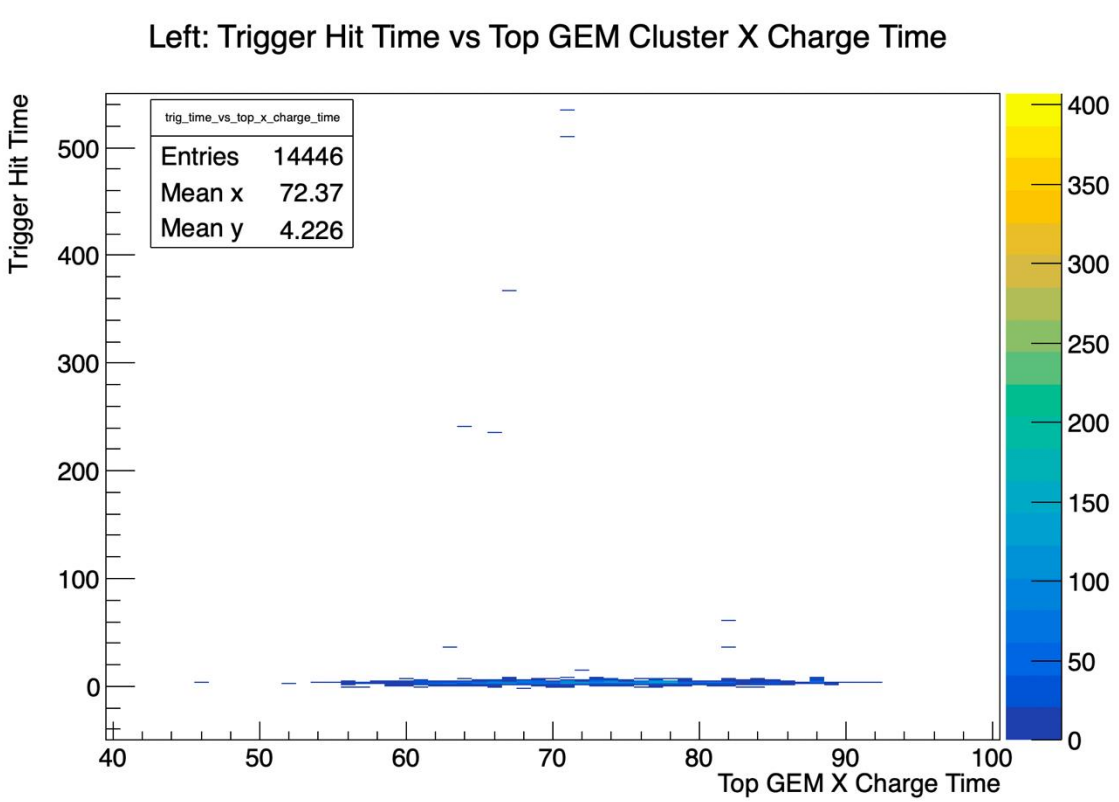
Single (one hit per detector per event) and On-Paddle (hit scintillators) tracks

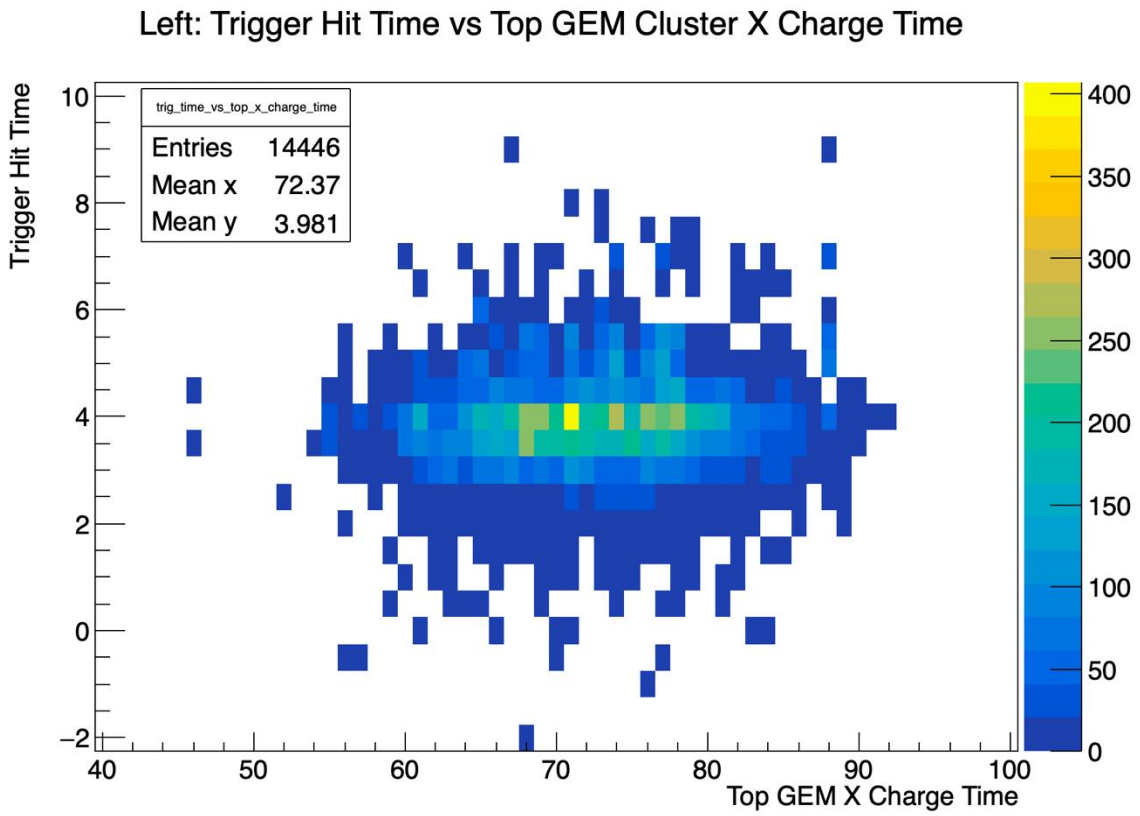
Correlation Between Trigger and GEM Cluster Times

Charge Time

Trigger vs Left Top GEM X axis

Run 3622 - Cosmic





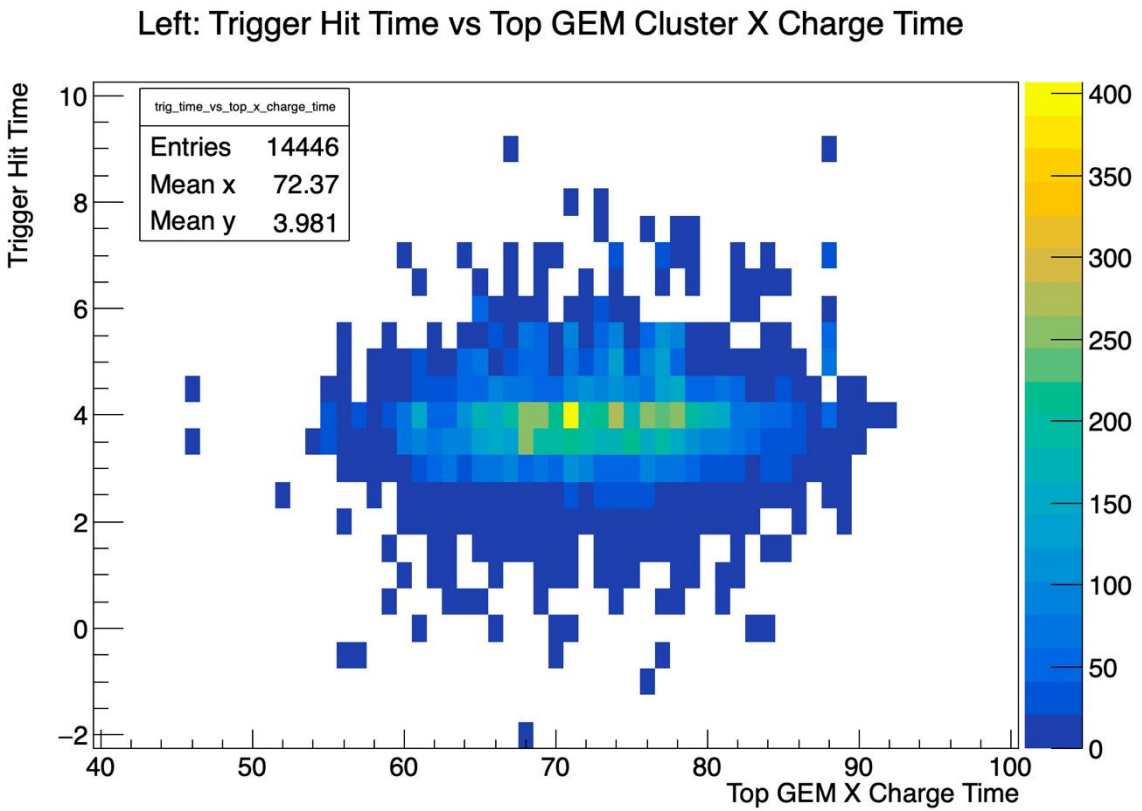
All tracks

Correlation Between Trigger and GEM Cluster Times

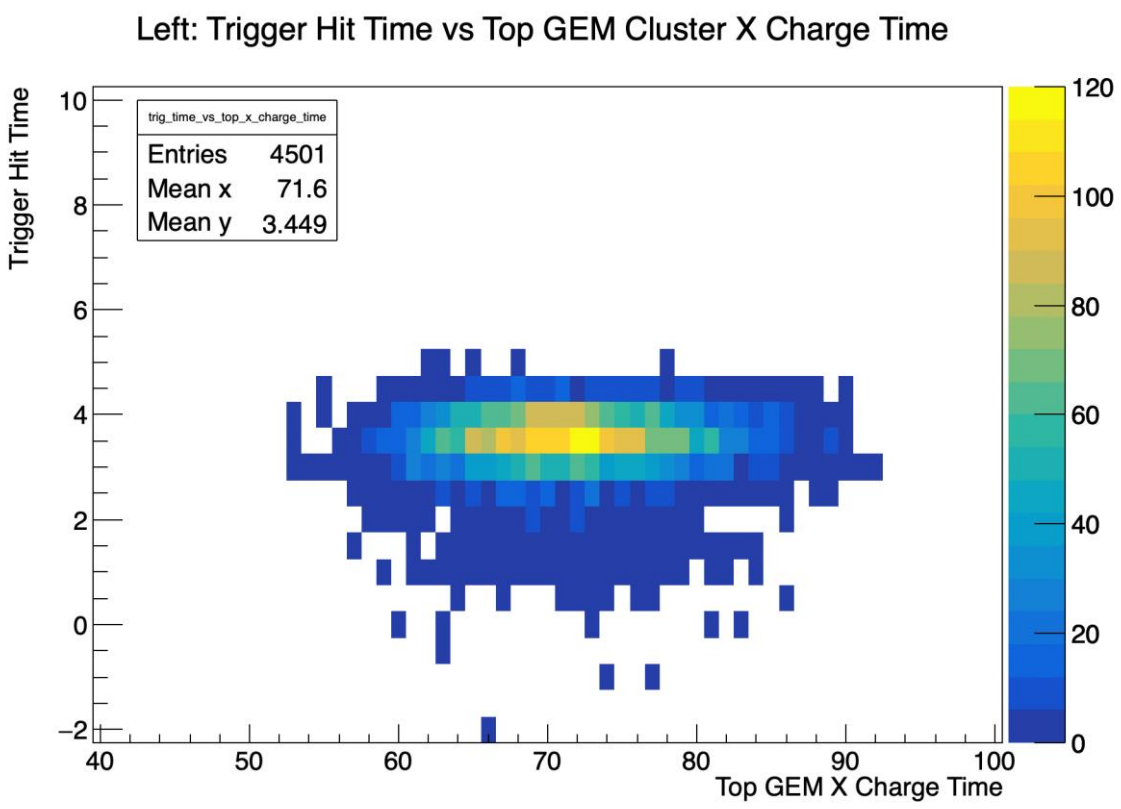
Charge Time

Trigger vs Left Top GEM X axis

Run 3622 - Cosmic



All tracks



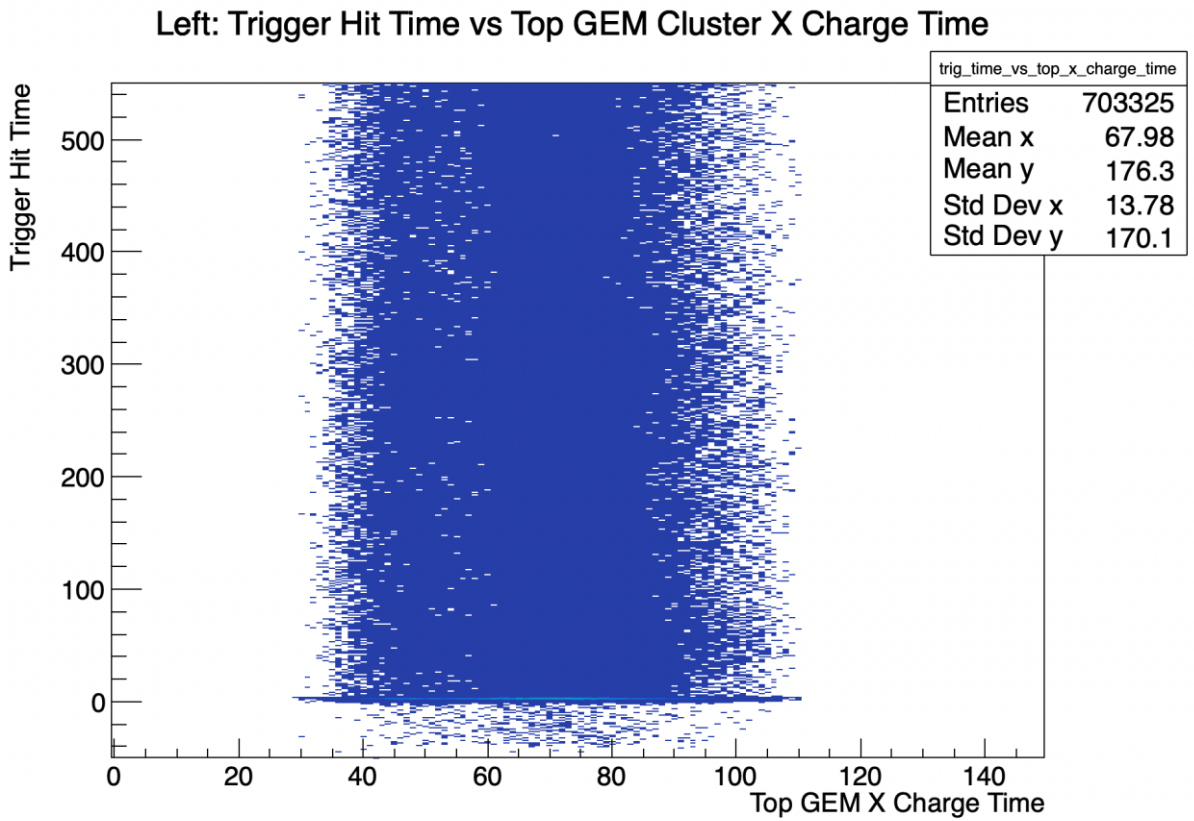
Single (one hit per detector per event) and On-Paddles (hits scintillators) tracks

Correlation Between Trigger and GEM Cluster Times

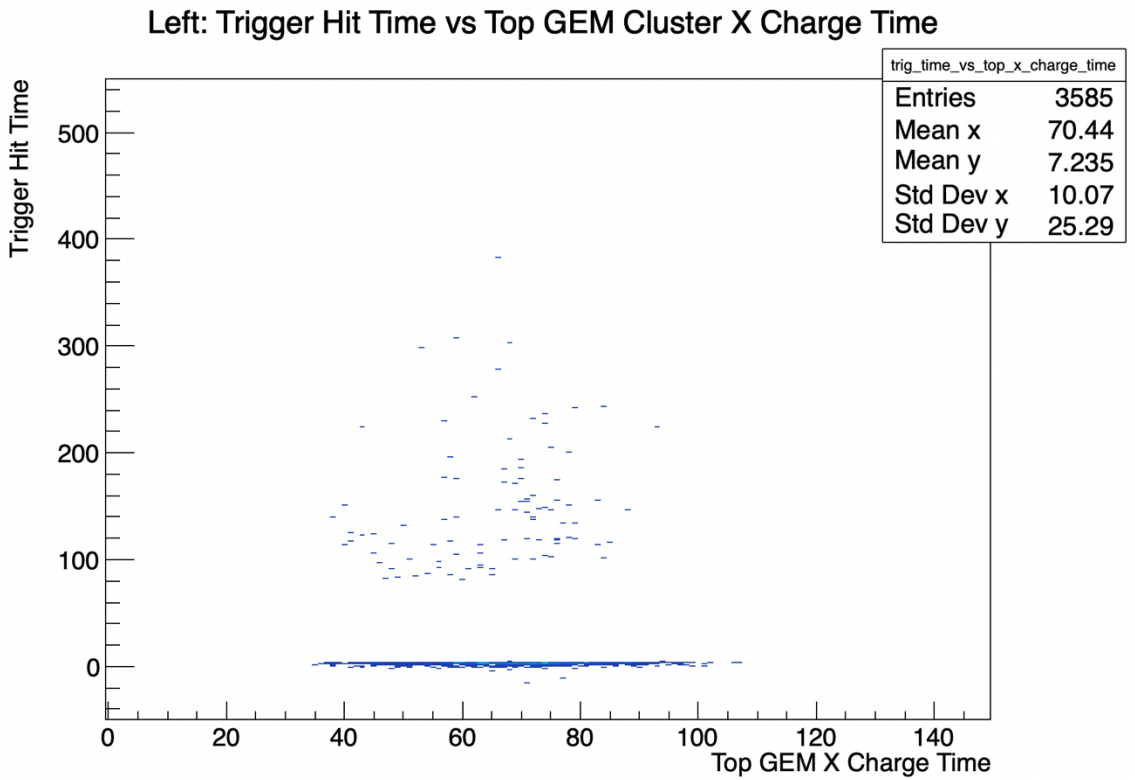
Charge Time

Trigger vs Left Top GEM X axis

Run 4010 - Elastic



All tracks



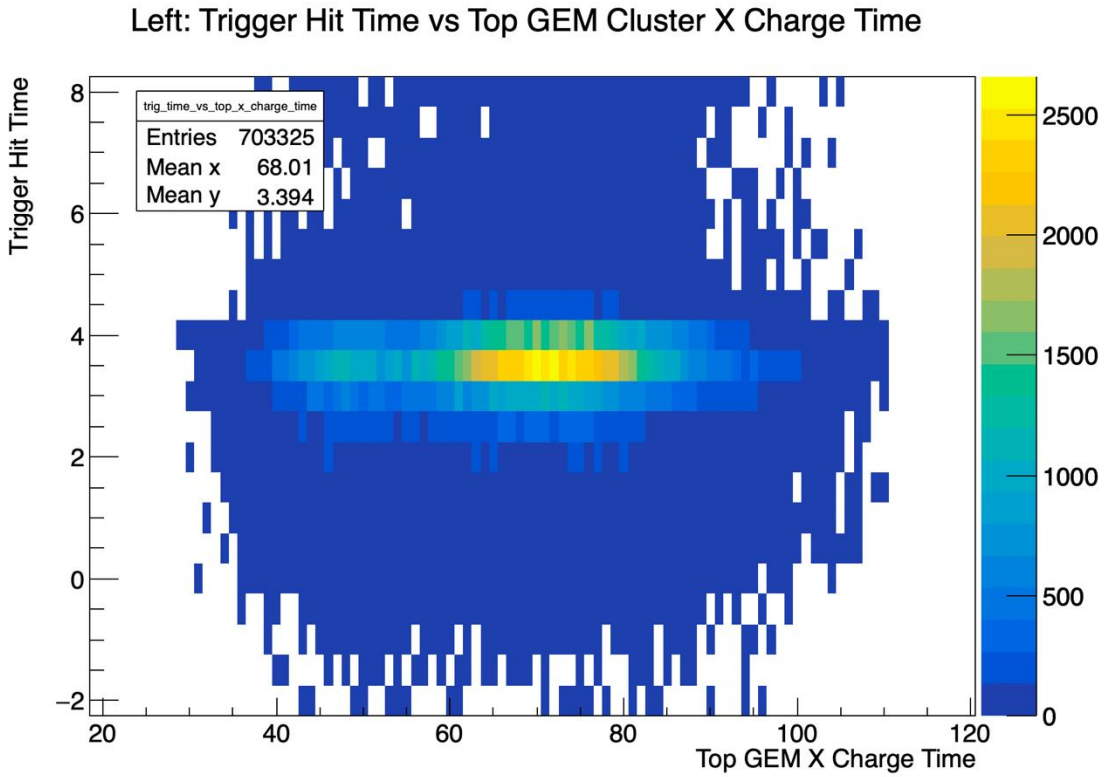
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Correlation Between Trigger and GEM Cluster Times

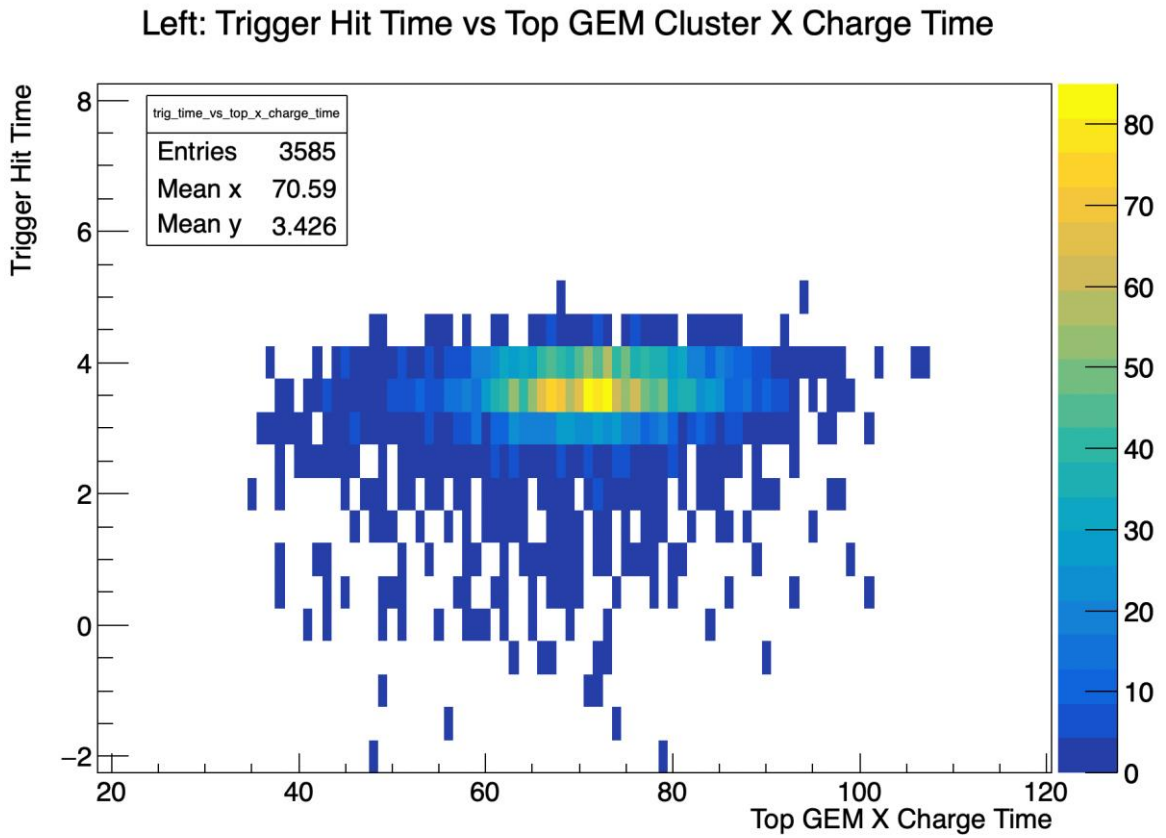
Charge Time

Trigger vs Left Top GEM X axis

Run 4010 - Elastic



All tracks



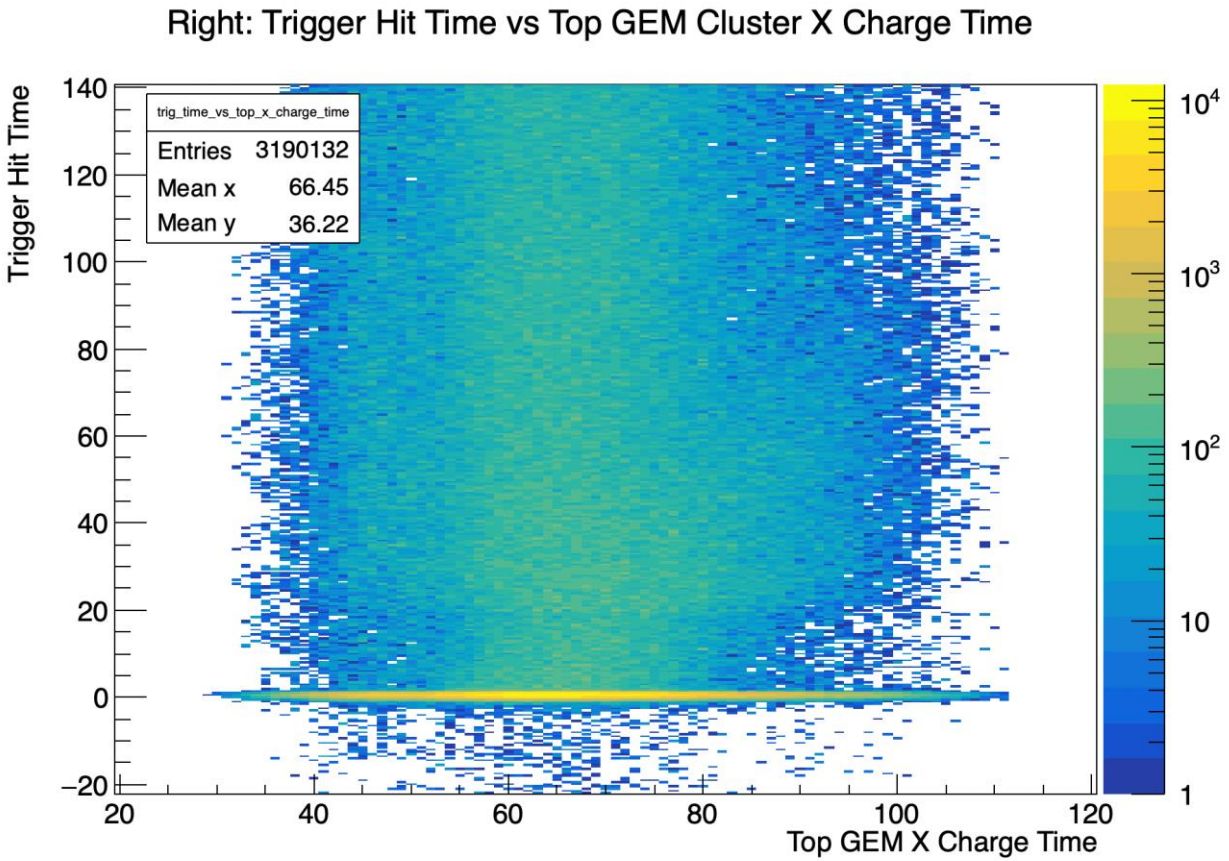
Single (one hit per detector per event) and On-Paddles (hits scintillators) tracks

Correlation Between Trigger and GEM Cluster Times

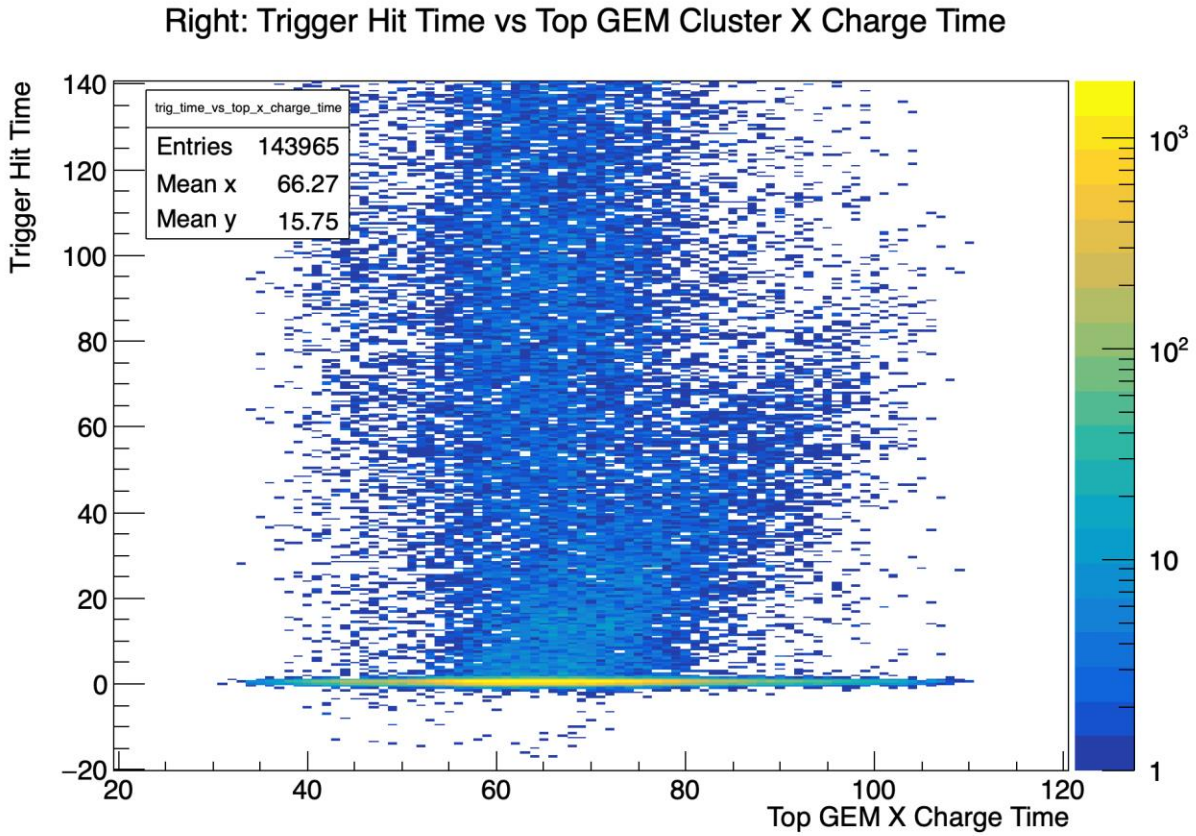
Charge Time

Trigger vs Right Top GEM X axis

Run 4001 - Elastic



All tracks



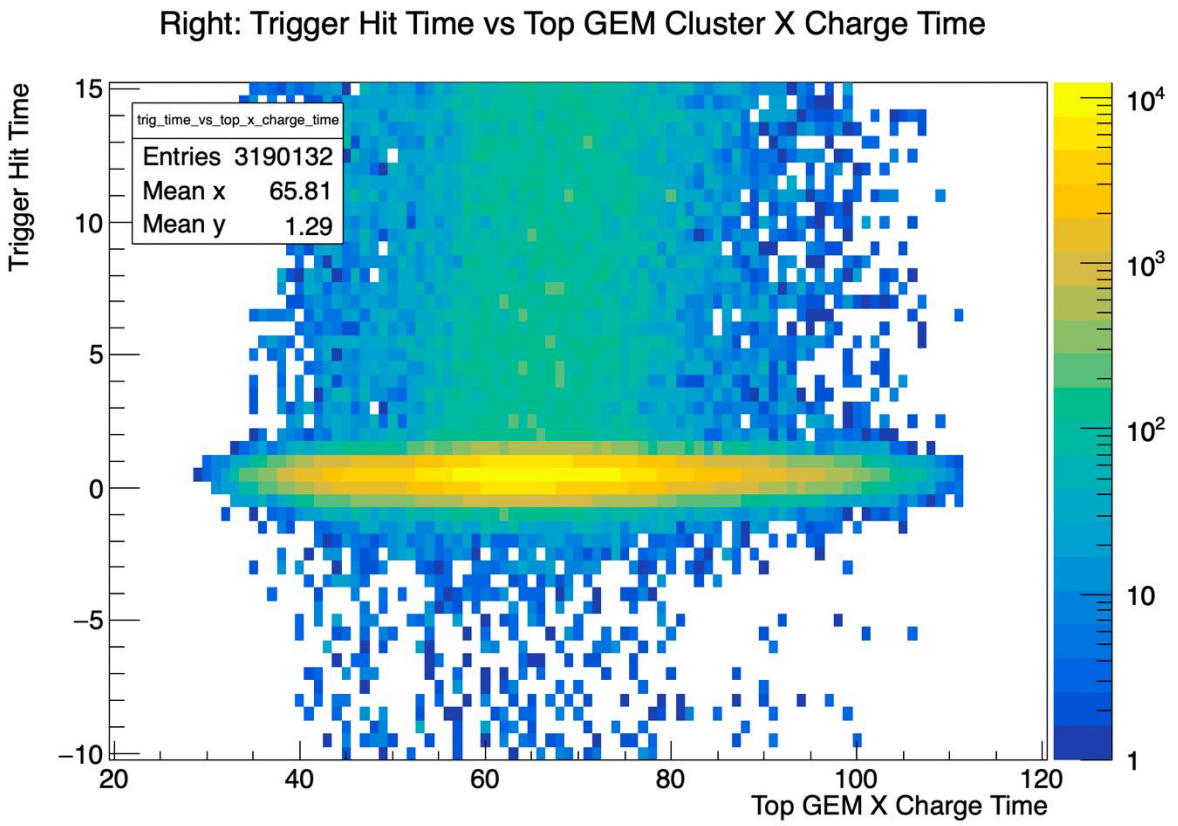
Best (closest to a trigger hit) and On-Paddles (hits scintillators) tracks

Correlation Between Trigger and GEM Cluster Times

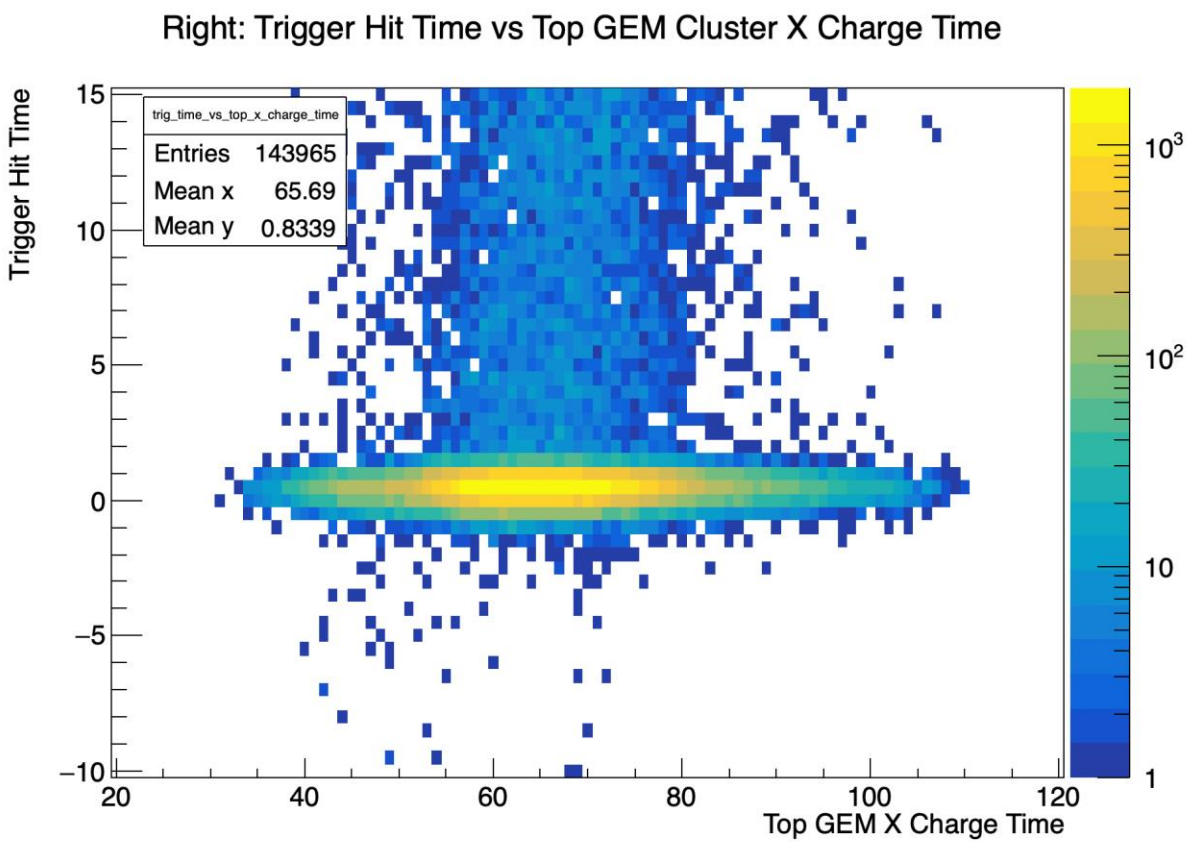
Charge Time

Trigger vs Right Top GEM X axis

Run 4001 - Elastic



All tracks



Best (closest to a trigger hit) and On-Paddles (hits scintillators) tracks

Conclusion

- Tested two rough methods of determining times of GEM clusters
  - No clear advantage between one method over the other
- Observed that many clusters identified by the GEMini code appeared noisy or difficult to interpret
- Clear correlation between cluster times in X and Y axis of individual GEMs
- Correlations between cluster times between GEMs and between GEMs and trigger hits is not clear

Next steps

1. Implement improved pedestal subtraction code (Evan)
2. Test and improve cluster identification code
   - ”Train” algorithm on good clusters which are part of good tracks
   - Consider implementing time-domain fit
3. Continue to analyze GEM clusters
   - Consider a smarter fitting or peak-finding method to determine cluster time
   - Can we see the effect of track angle on the cluster signal (time projection)?
   - Analyze correlation between GEM cluster times and trigger hit times again with improved time determination method and improved lower-level data processing
   - Maybe: Analyze amplitude or area of clusters
4. Implement new selections into track reconstruction code

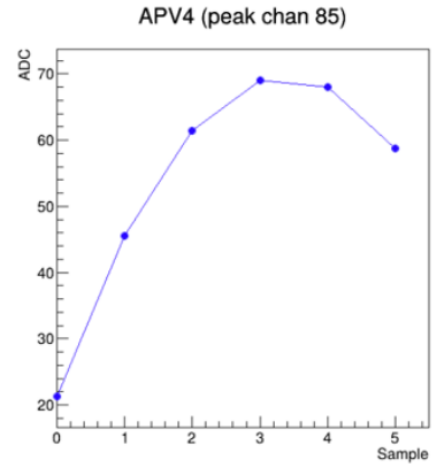
Thank you

Previous work

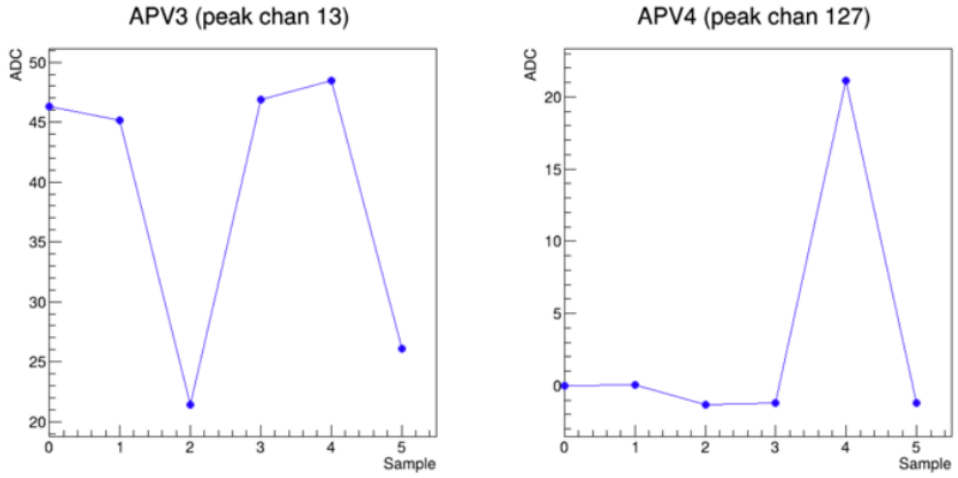
Co-op students have looked into timing of GEM data in simulation.

- Sophie looked at how the ADC value of the “peak” channel of the cluster changes over time samples
- Andy looked at using the GEM as a time-projection chamber to reconstruct track angles

Credit: Sophie Gelinas



(a)



(b)

Figure 5: ADC of peak channel vs. sample number for a single APV. (a) shows an x-axis APV, and (b) shows two y-axis APVs.